

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\E-32002 Intercooler.pvdb

Date of Analysis : May 21, 2026 5:06pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 850.000 mm ID
 Vessel Design Length, Tangent to Tangent 907.80 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 125 °C
 Channel Side Design Temperature 125 °C
 Shell Side Design Pressure 11.952 kgf/cm²
 Channel Side Design Pressure 9.232 kgf/cm²
 Shop Shell Side Test Pressure 25.982 kgf/cm²
 Shop Channel Side Test Pressure 16.724 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-214	...	...	K01807	
Tubesheet	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. COVER	9.232	0.00	13.000	3.1750	N/A	No
CH. FLG1	9.232	0.00	21.000	3.1750	N/A	No
CH. SHELL1	9.317	0.00	29.000	3.1750	N/A	No
SHELL	12.000	0.00	29.000	3.1750	N/A	No
CH SHELL2	9.317	0.00	29.000	3.1750	N/A	No
CH. FLG2	9.232	0.00	21.000	3.1750	N/A	No
CH. COVER2	9.232	0.00	13.000	3.1750	N/A	No

Liquid Level: 85.00 cm Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	6.0	5.5	70.0	55.0	47.1	...	1.00	1.00
Body Flg	18.5	12.5	83.0	75.0	72.7	...	1.00	1.00
Cylinder	88.4	70.0	12.0	12.0	6.0	...	1.00	1.00
Cylinder	815.5	723.3	12.0	12.0	6.8	...	1.00	1.00



Cylinder	889.3	70.0	12.0	12.0	6.0	...	1.00	1.00
Body Flg	902.3	12.5	83.0	75.0	63.3	...	1.00	1.00
Body Flg	907.8	5.5	70.0	55.0	47.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS Inlet Nozzl	WRC-297	0.324	Shell	Passed
SS INLET NOZZL	WRC-297	0.416	Shell	Passed
SS OUTLET NOZZ	WRC-297	0.418	Shell	Passed
TS OUT Nozzle	WRC-297	0.321	Shell	Passed

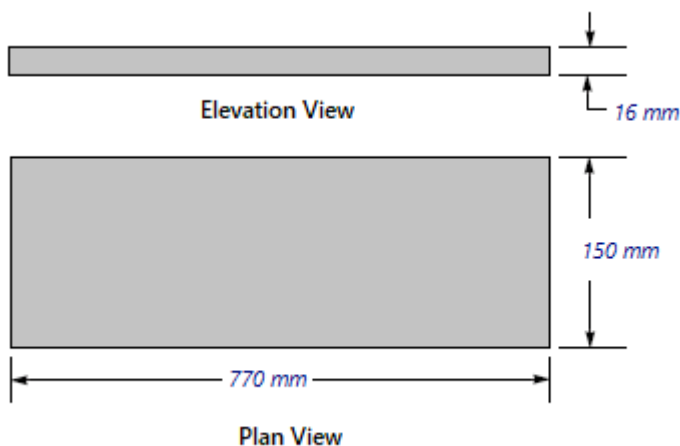
Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	680.000 mm
Wear Pad Width	225.000 mm
Wear Pad Thickness	12.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1460.000 mm

Baseplate Length	770.000 mm
Baseplate Thickness	16.000 mm
Baseplate Width	150.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	16.000 mm
Web Thickness	16.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	2

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	9105.71 kgf
Maximum Transverse Saddle Shear Load	1007.40 kgf
Maximum Longitudinal Saddle Shear Load	1098.48 kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	7994.69 kgf
Maximum Transverse Saddle Shear Load	332.44 kgf
Maximum Longitudinal Saddle Shear Load	62.90 kgf

Weights:

Fabricated - Bare w/o removable internals	9265.5 kg
---	-----------



Shop Test	- Fabricated + water (full)	13833.5	kg
Shipping	- Fab. + rem. intls.+ shipping app.	9406.6	kg
Erected	- Fab. + rem. intls.+ insul. (etc)	9406.6	kg
Empty	- Fab. + intls. + details + wghts.	9406.6	kg
Operating	- Empty + operating liquid (no ca)	12571.7	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS Inlet Nozzle	16	...	...	OK	6.02	OK	Passed
SS INLET NOZZLE	20	...	...	OK	6.83	OK	Passed
SS OUTLET NOZZL	20	...	...	OK	6.85	OK	Passed
TS OUT Nozzle	16	...	...	OK	6.00	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 20 Nozzle : SS OUTLET NOZZL [Shellside]
 Minimum MAWP Nozzles : 16 Nozzle : TS Inlet Nozzle [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	TS Inlet Nozzle	534.500	-90.000	521.716
40	SS INLET NOZZLE	7622.500	90.000	968.350
40	SS OUTLET NOZZL	1472.499	-90.000	968.350
50	TS OUT Nozzle	8543.500	90.000	521.716

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
TS Inlet Nozzle	10.000 in	40	WNF	10.750	9.271	373.05	12.00	284.53	300
TS OUT Nozzle	10.000 in	40	WNF	10.750	9.271	373.05	12.00	284.53	300
SS INLET NOZZLE	20.000 in	40	WNF	20.000	15.088	608.00	12.00	346.25	300
SS OUTLET NOZZL	20.000 in	40	WNF	20.000	15.088	608.00	12.00	346.25	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
TS Inlet No	SA-106 B	7.112	10.000	10.000	12.000	...
TS OUT Nozz	SA-106 B	7.112	10.000	10.000	12.000	...
SS INLET NO	SA-106 B	12.000	10.000	10.000	12.000	...
SS OUTLET N	SA-106 B	12.000	10.000	10.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
TS Inlet Nozzle	53.450	-90.0	250.00	0.00	CH. SHELL1
TS OUT Nozzle	854.350	90.0	250.00	0.00	CH SHELL2
SS INLET NOZZLE	762.250	90.0	250.00	0.00	SHELL
SS OUTLET NOZZL	147.250	-90.0	250.00	0.00	SHELL

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 70.0mm THK X 1020.0mm OD	SA-516 70
2	WELD NECK FLANGE: 83.0mm THK X 1020.0mm OD	SA-266 2
2	CYLINDER: 12.0mm THK X 850.0mm ID X 70.0cm	SA-516 70
1	CYLINDER: 12.0mm THK X 850.0mm ID X 723.3cm	SA-516 70
1	BLIND FLANGE: 70.0mm THK X 1020.0mm OD	SA-266 2
8	BAFFLES: 842.0mm DIA X 8.0mm THK	SA-516 70
2	TUBESHEET: 874.0mm X 38.0mm THK	SA-516 70
602	TUBES: 7309.0mm X 19.0mm DIA X 2.1mm THK	SA-214
2	SADDLE: 200.0mm X 120 DEG	SA-516 70
1	INSULATION: 7233mm X 45mm THK	
2	GASKET: 882.4mm OD X 856.4mm ID	...
56	BODY FLANGE BOLTS: 24.0mm DIA	SA-193 B7
112	NUTS FOR BODY FLANGE BOLTS: 24.0mm DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
SHELL	[8]	B	-23	-40	-29	0.695	12.000	1.00	No
Left Saddle	[24]	B	-23	-40	-29	0.695	12.000	1.00	No
RIGHT SADDLE	[24]	B	-23	-40	-29	0.695	12.000	1.00	No
SS INLET NOZZL	[1]	B	-23	-48	-29	0.414	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
SS OUTLET NOZZL	[1]	B	-23	-48	-29	0.416	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Tubesheet: SS	[13]	B	-23	-104	-29	0.349	12.000	1.00	No
Warmest MDMT:			-18	-40					
CH. COVER	[11]	B	-19	-25	-29	0.888	13.750	1.00	No
CH. FLG1	[11]	B	-23	-48	-29	0.449	12.000	1.00	No
CH. FLG2	[11]	B	-23	-48	-29	0.449	12.000	1.00	No
CH. COVER2	[11]	B	-19	-25	-29	0.888	13.750	1.00	No
CH. SHELL1	[8]	B	-23	-48	-29	0.446	12.000	1.00	No
CH SHELL2	[8]	B	-23	-48	-29	0.446	12.000	1.00	No
TS Inlet Nozzl	[1]	B	-8	-104	-29	0.206	8.112	1.00	No
Nozzle Flg	[5]	A	-18	-104					
TS OUT Nozzle	[1]	B	-8	-104	-29	0.204	8.112	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Tubesheet: CS	[14]	B	-23	-48	-29	0.356	12.000	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						
Warmest MDMT:			-8	-25					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-40.0		-28.9		Pass			
Channel/Tube		-29.0		-28.9		Pass			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	11.952	kgf/cm ²
Channel Side Design Pressure	9.232	kgf/cm ²
Shell Side Design Temperature	125.0	°C
Channel Side Design Temperature	125.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
-------------	------



0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y

Consider MAP New and Cold in Noz. Design N

Consider External Loads for Nozzle Des. Y

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No

Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	CH. COVER
Distance "FROM" to "TO"	5.5 cm
Flange Outside Diameter	1020 mm
Element Thickness	55 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	70 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.2317 kgf/cm ²
Design Temperature Internal Pressure	125 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²



Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2396.2	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	2/7	
Element From Node	20	
Element To Node	30	
Element Type	Flange	
Description	CH. FLG1	
Distance "FROM" to "TO"	12.5	cm
Flange Inside Diameter	850	mm
Element Thickness	75	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	83	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.2317	kgf/cm ²
Design Temperature Internal Pressure	125	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2294.2	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2273.9	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	3/7	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	CH. SHELL1	
Distance "FROM" to "TO"	70	cm
Inside Diameter	850	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.2317	kgf/cm ²
Design Temperature Internal Pressure	125	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2396.2	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	



UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	85 cm
Liquid Density	999.54 kg/m ³
Element From Node	30
Detail Type	Nozzle
Detail ID	TS Inlet Nozzle
Dist. from "FROM" Node / Offset dist	35 cm
Nozzle Diameter	10 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	-90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	SHELL
Distance "FROM" to "TO"	723.3 cm
Inside Diameter	850 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	11.952 kgf/cm ²
Design Temperature Internal Pressure	125 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	146 cm
Width of Saddle	200 mm
Height of Saddle at Bottom	680 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	225 mm
Thickness of Wear Plate	12 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.5
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm

Element From Node	40
Detail Type	Saddle
Detail ID	RIGHT SADDLE



Dist. from "FROM" Node / Offset dist	585.5	cm
Width of Saddle	200	mm
Height of Saddle at Bottom	680	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	225	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.5	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	85	cm
Liquid Density	562.74	kg/m ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	723.3	cm
Thickness of Insulation	45	mm
Density	0.00015	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	SS INLET NOZZLE	
Dist. from "FROM" Node / Offset dist	670	cm
Nozzle Diameter	20	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	SS OUTLET NOZZL	
Dist. from "FROM" Node / Offset dist	55	cm
Nozzle Diameter	20	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	CH SHELL2
Distance "FROM" to "TO"	70 cm
Inside Diameter	850 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.2317 kgf/cm ²



Design Temperature Internal Pressure 125 °C
Design External Pressure 0 kgf/cm²
Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Weld is pre-Heated No

Element From Node 50
Detail Type Liquid
Detail ID Liquid: 50
Dist. from "FROM" Node / Offset dist 0 cm
Height/Length of Liquid 85 cm
Liquid Density 999.54 kg/m³

Element From Node 50
Detail Type Nozzle
Detail ID TS OUT Nozzle
Dist. from "FROM" Node / Offset dist 35 cm
Nozzle Diameter 10 in.
Nozzle Schedule 40
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-106 B

Element# 6/7
Element From Node 60
Element To Node 70
Element Type Flange
Description CH. FLG2
Distance "FROM" to "TO" 12.5 cm
Flange Inside Diameter 850 mm
Element Thickness 75 mm
Internal Corrosion Allowance 3.175 mm
Nominal Thickness 83 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 9.2317 kgf/cm²
Design Temperature Internal Pressure 125 °C
Design External Pressure 0 kgf/cm²
Design Temperature External Pressure 0 °C
Effective Diameter Multiplier 1.2
Material Name SA-266 2
Allowable Stress, Ambient 1407.1 kgf/cm²
Allowable Stress, Operating 1407.1 kgf/cm²
Allowable Stress, Hydrotest 2294.2 kgf/cm²
Material Density 0.00775 kg/cm³
P Number Thickness 32.004 mm
Yield Stress, Operating 2273.9 kgf/cm²
UCS-66 Chart Curve Designation B
External Pressure Chart Name CS-2
UNS Number K03506
Product Form Forgings
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No

Element# 7/7
Element From Node 70
Element To Node 80
Element Type Flange
Description CH. COVER2
Distance "FROM" to "TO" 5.5 cm
Flange Outside Diameter 1020 mm



Element Thickness	55	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	70	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.2317	kgf/cm ²
Design Temperature Internal Pressure	125	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

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XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
CH. COVER		5.95	...	5.5	...
CH. FLG1		18.45	...	12.5	...
CH. SHELL1		88.45	...	70	...
SHELL		815.55	...	723.3	...
CH SHELL2		889.35	...	70	...
CH. FLG2		902.3	...	12.5	...
CH. COVER2		907.8	...	5.5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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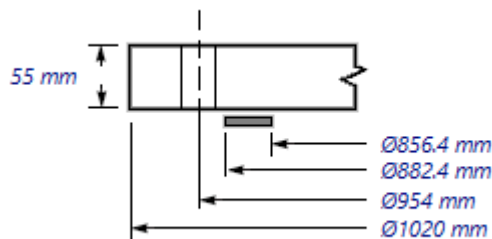


Flange Input Data Values

Description: FLANGE :

CH. COVER

Flange Type		Blind	
Design Pressure	P	9.23	kgf/cm ²
Design Temperature		125	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1020.000	mm
Flange Thickness	t	55.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	2997.079	mm
Diameter of Bolt Circle	C	954.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	962.400	mm
Flange Face Inside Diameter	Fid	850.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	882.400	mm
Gasket Inside Diameter	Gi	856.400	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	869.551 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 869.5509^2 \cdot 9.232$
 $= 54822.812 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 869.5509 \cdot 3.75 \cdot 9.23$$

$$= 12151.526 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(54823 + 12152 + 0, 0)$$

$$= 66974.344 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 534.33 \cdot 6.4246 \cdot \pi \cdot 869.551 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 93777.727 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((66974 + 0 + 4 \cdot 0/869.55)/1754, 93778/1754)$$

$$= 53.471 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	53.471	87.569	
Radial Distance between Bolts and Edge:	29.000	33.000	
Circ. Spacing between the Bolts:	57.500	106.814	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 87.569 \cdot 1753.82 / (534.33 \cdot \pi (882.4 + 856.4))$$

$$= 5.262 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 87.5694$$

$$= 153580.77 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 66974.34 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (954.0 - 869.5509) / 2$$

$$= 42.2246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 66974 \cdot 42.2246 \cdot 1.0 / (51.825^2 \cdot 869.5509) + 0.3 \cdot 1.0 \cdot 9.23 \cdot 869.5509^2 / 51.825^2$$

$$= 1009.74 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 153581 \cdot 42.2246 \cdot 1.0 / (51.825^2 \cdot 869.5509)$$

$$= 527.57 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 66974 / 87.5694$$

$$= 764.82 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (93778 / 87.5694)$$



= 1070.90 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1009.74	1407.13	527.57 1407.13
Bolting	764.82	1753.82	1070.90 1753.82

Reqd. Blind Flange Thickness at Center 47.076 mm
 Reqd. Blind Flange Thickness at Gasket 34.908 mm

Estimated M.A.W.P. (Operating) 13 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 348.3 kg
 Estimated Unfinished Weight of Forging at given Thk 348.3 kg

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.888, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -19 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -25 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	896.53	1407.13	468.42 1407.13
Bolting	764.82	1753.82	1070.90 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 43.917 mm

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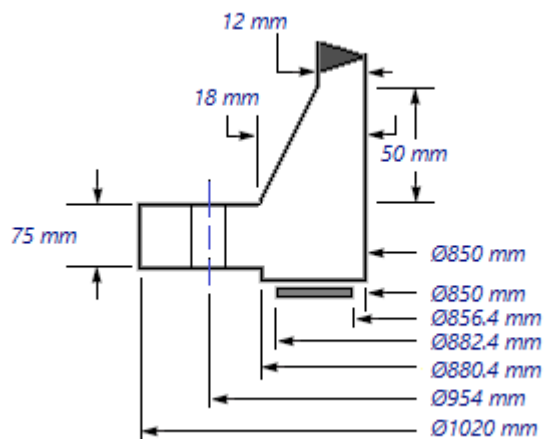


Flange Input Data Values

Description: FLANGE :

CH. FLG1

Flange Type	Integral Weld Neck		
Design Pressure	P	9.23	kgf/cm ²
Design Temperature		125	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	850.000	mm
Flange Outside Diameter	A	1020.000	mm
Flange Thickness	t	75.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	18.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	954.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	880.400	mm
Flange Face Inside Diameter	Fid	850.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	882.400	mm
Gasket Inside Diameter	Gi	856.400	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	856.4000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	534.33	kgf/cm ²





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Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (9.23 \cdot (850.0/2 + 3.175)) / (1407.13 \cdot 1.0 - 0.6 \cdot 9.23) + Ca \\ &= 5.9952 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 8.825) / (428.175 + 0.6 \cdot 8.825) \\ &= 28.648 \text{ kgf/cm}^2 \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 12.0) / (425.0 + 0.6 \cdot 12.0) \\ &= 39.069 \text{ kgf/cm}^2 \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	856.350 mm
Corroded Large Hub,	g1Cor = g1-ci	14.825 mm
Corroded Small Hub,	g0Cor = go-ci	8.825 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	34.000 mm
Gasket Contact Width,	N = (Fod - Gi)/2	12.000 mm
Basic Gasket Width,	bo = N/2	6.000 mm
Effective Gasket Width,	b = bo	6.000 mm
Gasket Reaction Diameter,	G = (Go + Gi)/2	868.400 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 868.4001^2 \cdot 9.232 \\ &= 54677.801 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P \\ &= 2 \cdot 6.0 \cdot \pi \cdot 868.4001 \cdot 3.75 \cdot 9.23 \\ &\quad + 2 \cdot 856.4 \cdot 5.0 \cdot 3.75 \cdot 9.2317 \\ &= 14298.244 \text{ kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 856.35^2 \cdot 9.2317 \\ &= 53170.895 \text{ kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 54678 - 53171 \\ &= 1506.907 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(54678 + 14298 + 0, 0) \\ &= 68976.047 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 534.33 \cdot 6.0 \cdot \pi \cdot 868.4 + 534.33 \cdot 5.0 \cdot 856.4 \\ &= 110344.727 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((68976 + 0 + 4 \cdot 0 / 868.4) / 1754, 110345 / 1754) \\ &= 62.917 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t/(m + 0.5) \\ &= 2 \cdot 24.0 + 6 \cdot 75.0 / (3.75 + 0.5) \\ &= 153.882 \text{ mm} \end{aligned}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 954.0 \cdot \sin(\pi / 28)$$

$$= 106.814 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{106.814 / (2 \cdot 24.0 + 75.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	62.917	87.569	
Radial Distance between Hub and Bolts:	33.500	34.000	
Radial Distance between Bolts and Edge:	29.000	33.000	
Circ. Spacing between the Bolts:	57.500	106.814	153.882

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 87.569 \cdot 1753.82 / (534.33 \cdot \pi(880.4 + 856.4))$$

$$= 5.268 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 87.5694$$

$$= 153580.77 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 68976 - 54678$$

$$= 14298.25 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (954.0 - 868.4001) / 2$$

$$= 42.8000 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((954.0 - 856.35) / 2 + 42.8)$$

$$= 45.8125 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (954.0 - 856.35 - 14.825) / 2$$

$$= 41.4125 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	53171.	41.4125	1.0000	2202.
Face Pressure, Mt	1507.	45.8125	1.0000	69.
Gasket Load, Mg	14298.	42.8000	1.0000	612.
Gasket Seating, Matm	153581.	42.8000	1.0000	6573.

Total Moment for Operation, Mo 2883. kgf-m.
 Total Moment for Gasket seating, Mg 6573. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 86.933 mm
 Hub Ratio, h/h0 = HL / H0 0.575



Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.680

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.824
 Factor V per 4.16.5 0.259
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1020.0/856.35
 = 1.191

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 = $(0.66845 + (5.7169(1.191^2 \cdot \log_{10}(1.191))) / (1.191^2 - 1)) / (1.191 - 1)$
 = 11.196

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 = $(1.191^2(1 + 8.55246 \cdot \log_{10}(1.191)) - 1) / ((1.0472 + 1.9448 \cdot 1.191^2) \cdot (1.191 - 1))$
 = 1.843

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 = $(1.191^2(1 + 8.55246 \cdot \log_{10}(1.191)) - 1) / (1.36136(1.191^2 - 1) \cdot (1.191 - 1))$
 = 12.303

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
 = $(1.191^2 + 1) / (1.191^2 - 1)$
 = 5.776

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$
 = $12.303 \cdot 8.825^2 \cdot 86.933 / 0.259$
 = 322144.156 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.824/86.933
 = 0.009 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
 = $(75.0 \cdot 0.009 + 1) / 1.843 + 75.0^3/322144.156$
 = 2.238

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 2883/856.35) / (2.2379 \cdot 14.825^2)$
 = 684.46 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 6573/856.35) / (2.2379 \cdot 14.825^2)$
 = 1560.60 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 = $(1.9474 \cdot 2883/856.35) / (2.2379 \cdot 75.0^2)$
 = 52.08 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 = $(1.9474 \cdot 6573/856.35) / (2.2379 \cdot 75.0^2)$
 = 118.74 kgf/cm²

Tangential Flange Stress, Operating [STo]:



$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (11.196 \cdot 2883 / (75.0^2 \cdot 856.35)) - 5.7764 \cdot 52$$

$$= 369.25 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (11.196 \cdot 6573 / (75.0^2 \cdot 856.35)) - 5.7764 \cdot 119$$

$$= 841.91 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (684 + \max(52, 369)) / 2$$

$$= 526.85 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRA, STa)) / 2$$

$$= (1561 + \max(119, 842)) / 2$$

$$= 1201.25 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 68976 / 87.5694$$

$$= 787.67 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (110345 / 87.5694)$$

$$= 1260.08 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	684.46	2110.70	1560.60	2110.70
Radial Flange	52.08	1407.13	118.74	1407.13
Tangential Flange	369.25	1407.13	841.91	1407.13
Maximum Average	526.85	1407.13	1201.25	1407.13
Bolting	787.67	1753.82	1260.08	1753.82

Minimum Required Flange Thickness [Rigidity] 72.720 mm

Estimated M.A.W.P. (Operating) 21 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 160.9 kg
 Estimated Unfinished Weight of Forging at given Thk 241.9 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6573.3 / 1.0 \cdot 100000.023 \cdot 0.259 / (2.11 \cdot 2067026 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2882.9 / 1.0 \cdot 100000.023 \cdot 0.259 / (2.11 \cdot 2004811 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 0.452 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6573.3 / 1.0 \cdot 100000.023 \cdot 0.259 / (2.238 \cdot 2067026 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 0.943 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2882.9 / 1.0 \cdot 100000.023 \cdot 0.259 / (2.238 \cdot 2004811 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 0.426 \text{ (should be } \leq 1)$$



Minimum Design Metal Temperature Results:

Thickness Ratio = 0.449, Temperature Reduction per Fig. UCS 66.1 = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	652.62	2110.70	1442.83	2110.70
Radial Flange	69.14	1407.13	152.85	1407.13
Tangential Flange	285.05	1407.13	630.20	1407.13
Maximum Average	468.84	1407.13	1036.52	1407.13
Bolting	787.67	1753.82	1260.08	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 59.080 mm

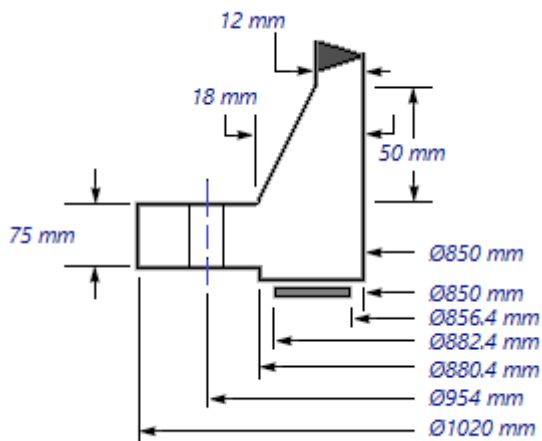
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Flange Input Data Values

Description: New Flange :

CH. FLG2

Flange Type	Integral Weld Neck		
Design Pressure	P	9.23	kgf/cm ²
Design Temperature		125	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	850.000	mm
Flange Outside Diameter	A	1020.000	mm
Flange Thickness	t	75.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	18.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	954.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	880.400	mm
Flange Face Inside Diameter	Fid	850.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	882.400	mm
Gasket Inside Diameter	Gi	856.400	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.23 \cdot (850.0/2 + 3.175)) / (1407.13 \cdot 1.0 - 0.6 \cdot 9.23) + Ca$$

$$= 5.9952 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 8.825) / (428.175 + 0.6 \cdot 8.825)$$

$$= 28.648 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 12.0) / (425.0 + 0.6 \cdot 12.0)$$

$$= 39.069 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	856.350 mm
Corroded Large Hub,	g1Cor = g1-ci	14.825 mm
Corroded Small Hub,	g0Cor = go-ci	8.825 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	34.000 mm

Gasket Contact Width,	N = (Fod - Gi)/2	12.000 mm
Basic Gasket Width,	bo = N/2	6.000 mm
Effective Gasket Width,	b = bo	6.000 mm
Gasket Reaction Diameter,	G = (Go + Gi)/2	868.400 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 868.4001^2 \cdot 9.232$$

$$= 54677.801 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.0 \cdot \pi \cdot 868.4001 \cdot 3.75 \cdot 9.23$$

$$= 11333.485 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 856.35^2 \cdot 9.2317$$

$$= 53170.895 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 54678 - 53171$$

$$= 1506.907 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(54678 + 11333 + 0, 0)$$

$$= 66011.289 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 534.33 \cdot 6.0 \cdot \pi \cdot 868.4 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 87464.609 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((66011 + 0 + 4 \cdot 0 / 868.4) / 1754, 87465 / 1754)$$

$$= 49.871 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 24.0 + 6 \cdot 75.0 / (3.75 + 0.5)$$

$$= 153.882 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 954.0 \cdot \sin(\pi / 28)$$

$$= 106.814 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:



$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{106.814}{2 \cdot 24.0 + 75.0}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	49.871	87.569	
Radial Distance between Hub and Bolts:	33.500	34.000	
Radial Distance between Bolts and Edge:	29.000	33.000	
Circ. Spacing between the Bolts:	57.500	106.814	153.882

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 87.569 \cdot 1753.82 / (534.33 \cdot \pi(880.4 + 856.4))$$

$$= 5.268 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 1753.82 (49.871 + 87.5694) / 2$$

$$= 120522.70 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 66011 - 54678$$

$$= 11333.49 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (954.0 - 868.4001) / 2$$

$$= 42.8000 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((954.0 - 856.35) / 2 + 42.8)$$

$$= 45.8125 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (954.0 - 856.35 - 14.825) / 2$$

$$= 41.4125 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	53171.	41.4125	1.0000	2202.
Face Pressure, Mt	1507.	45.8125	1.0000	69.
Gasket Load, Mg	11333.	42.8000	1.0000	485.
Gasket Seating, Matm	120523.	42.8000	1.0000	5158.

Total Moment for Operation, Mo 2756. kgf-m.
 Total Moment for Gasket seating, Mg 5158. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 86.933 mm
 Hub Ratio, h/h0 = HL / H0 0.575
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.680

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.824
 Factor V per 4.16.5 0.259
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1020.0/856.35$$

$$= 1.191$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.191^2 \cdot \log_{10}(1.191))) / (1.191^2 - 1)) / (1.191 - 1)$$

$$= 11.196$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.191^2(1 + 8.55246 \cdot \log_{10}(1.191)) - 1) / ((1.0472 + 1.9448 \cdot 1.191^2) \cdot (1.191 - 1))$$

$$= 1.843$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.191^2(1 + 8.55246 \cdot \log_{10}(1.191)) - 1) / (1.36136(1.191^2 - 1) \cdot (1.191 - 1))$$

$$= 12.303$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.191^2 + 1) / (1.191^2 - 1)$$

$$= 5.776$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 12.303 \cdot 8.825^2 \cdot 86.933 / 0.259$$

$$= 322144.156 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.824/86.933$$

$$= 0.009 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (75.0 \cdot 0.009 + 1) / 1.843 + 75.0^3/322144.156$$

$$= 2.238$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 2756/856.35) / (2.2379 \cdot 14.825^2)$$

$$= 654.33 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 5158/856.35) / (2.2379 \cdot 14.825^2)$$

$$= 1224.68 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.9474 \cdot 2756/856.35) / (2.2379 \cdot 75.0^2)$$

$$= 49.79 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.9474 \cdot 5158/856.35) / (2.2379 \cdot 75.0^2)$$

$$= 93.18 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (11.196 \cdot 2756 / (75.0^2 \cdot 856.35)) - 5.7764 \cdot 50$$

$$= 353.00 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$



$$= (11.196 \cdot 5158 / (75.0^2 \cdot 856.35)) - 5.7764 \cdot 93$$

$$= 660.69 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (654 + \max(50, 353)) / 2$$

$$= 503.66 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1225 + \max(93, 661)) / 2$$

$$= 942.69 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 66011 / 87.5694$$

$$= 753.82 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (87465 / 87.5694)$$

$$= 998.80 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	654.33	2110.70	1224.68	2110.70
Radial Flange	49.79	1407.13	93.18	1407.13
Tangential Flange	353.00	1407.13	660.69	1407.13
Maximum Average	503.66	1407.13	942.69	1407.13
Bolting	753.82	1753.82	998.80	1753.82

Minimum Required Flange Thickness [Rigidity] 63.348 mm

Estimated M.A.W.P. (Operating) 21 kgf/cm²

Estimated Finished Weight of Flange at given Thk. 160.9 kg

Estimated Unfinished Weight of Forging at given Thk 241.9 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5158.4 / 1.0 \cdot 100000.023 \cdot 0.259 / (1.658 \cdot 2067026 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 0.999 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2756.0 / 1.0 \cdot 100000.023 \cdot 0.259 / (1.658 \cdot 2004811 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 0.550 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5158.4 / 1.0 \cdot 100000.023 \cdot 0.259 / (2.238 \cdot 2067026 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 0.740 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2756.0 / 1.0 \cdot 100000.023 \cdot 0.259 / (2.238 \cdot 2004811 \cdot 8.825^2 \cdot 86.933 \cdot 0.3)$$

$$= 0.407 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.449, Temperature Reduction per Fig. UCS 66.1 = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C



Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	624.77	2110.70	1132.27	2110.70
Radial Flange	66.19	1407.13	119.95	1407.13
Tangential Flange	272.89	1407.13	494.55	1407.13
Maximum Average	448.83	1407.13	813.41	1407.13
Bolting	753.82	1753.82	998.80	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 46.736 mm

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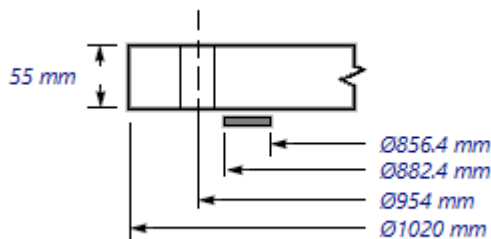


Flange Input Data Values

Description: New Flange :

CH. COVER2

Flange Type		Blind	
Design Pressure	P	9.23	kgf/cm ²
Design Temperature		125	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1020.000	mm
Flange Thickness	t	55.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	2997.079	mm
Diameter of Bolt Circle	C	954.000	mm
Nominal Bolt Diameter	a	24.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	962.400	mm
Flange Face Inside Diameter	Fid	850.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	882.400	mm
Gasket Inside Diameter	Gi	856.400	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	869.551 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 869.5509^2 \cdot 9.232$
 $= 54822.812 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 869.5509 \cdot 3.75 \cdot 9.23$$

$$= 12151.526 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(54823 + 12152 + 0, 0)$$

$$= 66974.344 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 534.33 \cdot 6.4246 \cdot \pi \cdot 869.551 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 93777.727 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((66974 + 0 + 4 \cdot 0/869.55)/1754, 93778/1754)$$

$$= 53.471 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	53.471	87.569	
Radial Distance between Bolts and Edge:	29.000	33.000	
Circ. Spacing between the Bolts:	57.500	106.814	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 87.569 \cdot 1753.82 / (534.33 \cdot \pi (882.4 + 856.4))$$

$$= 5.262 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 1753.82(53.4706 + 87.5694)/2$$

$$= 123679.24 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 66974.34 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (954.0 - 869.5509)/2$$

$$= 42.2246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 66974 \cdot 42.2246 \cdot 1.0 / (51.825^2 \cdot 869.5509) + 0.3 \cdot 1.0 \cdot 9.23 \cdot 869.5509^2 / 51.825^2$$

$$= 1009.74 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 123679 \cdot 42.2246 \cdot 1.0 / (51.825^2 \cdot 869.5509)$$

$$= 424.86 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 66974/87.5694$$

$$= 764.82 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (93778/87.5694)$$



= 1070.90 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1009.74	1407.13	424.86 1407.13
Bolting	764.82	1753.82	1070.90 1753.82

Reqd. Blind Flange Thickness at Center 47.076 mm
 Reqd. Blind Flange Thickness at Gasket 31.652 mm

Estimated M.A.W.P. (Operating) 13 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 348.3 kg
 Estimated Unfinished Weight of Forging at given Thk 348.3 kg

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.888, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -19 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -25 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	896.53	1407.13	377.22 1407.13
Bolting	764.82	1753.82	1070.90 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 43.917 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
CH. COVER		9.2317	70	3.175	1020	1407.1	1
CH. FLG1		9.2317	83	3.175	850	1407.1	2
CH. SHELL1		9.317	12	3.175	850	1407.1	3
SHELL		12	12	3.175	850	1407.1	4
CH SHELL2		9.317	12	3.175	850	1407.1	5
CH. FLG2		9.2317	83	3.175	850	1407.1	6
CH. COVER2		9.2317	70	3.175	1020	1407.1	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
CH. COVER		9.23171	12.8648	14.4895	55	47.0763	1
CH. FLG1		9.23171	20.5551	20.5551	75	72.7202	2
CH. SHELL1		9.23171	28.5625	39.069	12	6.02136	3
SHELL		11.9524	28.5998	39.069	12	6.84536	4
CH SHELL2		9.23171	28.5625	39.069	12	6.02136	5
CH. FLG2		9.23171	21.4783	21.4783	75	63.3476	6
CH. COVER2		9.23171	12.8648	14.4895	55	47.0763	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

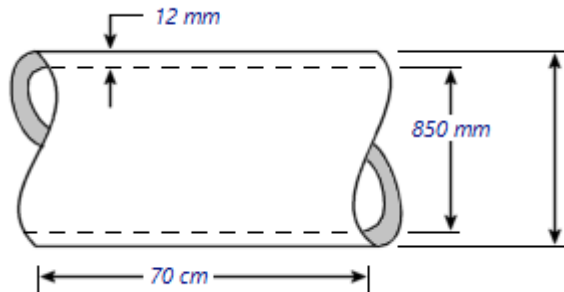
Shell Side MAWP = 20 kgf/cm²
 Shell Side MAPnc = 34 kgf/cm²
 Channel Side MAWP = 13 kgf/cm²
 Channel Side MAPnc = 14 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. SHELL1

Cylindrical Shell Design Information:					
Design Pressure	P	9.317 kgf/cm²	Int. Design Temperature	125 °C	
Material	SA-516 70		Ext. Design Temperature	0 °C	
Length	70.000 cm		External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	850.000 mm	Actual Stress	457.6 kgf/cm²	
Spec. Min. Thk t _e , or s _a	12.0000 mm		Spec. Nominal Thk.	12.000 mm	
Surface Area	19220.3 cm²		SG of Contents	1.000	
Element Volume	397216.1 cm³		Weight of Contents	397.0 kgf	
Empty Weight	176.3 kgf		Operating Weight	573.3 kgf	
			Comp. All. Ope. Stress	1329.8 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.317 \cdot 428.175) / (1407.13 \cdot 1.0 - 0.6 \cdot 9.317)$$

$$= 2.8464 + 3.1750 = 6.0214 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.085 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 8.825) / (428.175 + 0.6 \cdot 8.825)$$

$$= 29 - 0.1 = 29 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 12.0) / (425.0 + 0.6 \cdot 12.0)$$

$$= 39 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (9.317 \cdot (428.175 + 0.6 \cdot 8.825)) / (1.0 \cdot 8.825)$$

$$= 457.636 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.392 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 3.936, c = 3.175 mm, E* = 1.0

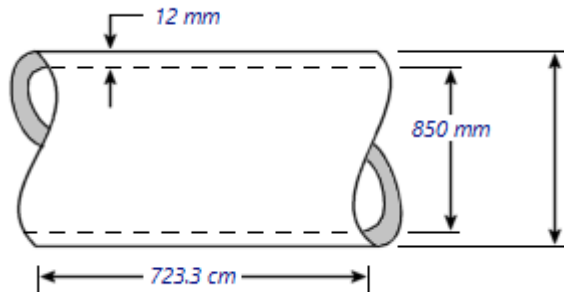
Thickness Ratio = $tr * E^* / (tg - c) = 0.446$, Temp. Reduction = 41 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.865 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.446	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	12.000 kgf/cm²	Int. Design Temperature	125 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		723.300 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	850.000 mm	Actual Stress	589.4 kgf/cm²	
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.	12.000 mm	
Surface Area		198600.3 cm²	SG of Contents	0.563	
Element Volume		2863308.5 cm³	Weight of Contents	1611.3 kgf	
Empty Weight		1821.7 kgf	Operating Weight	3433.0 kgf	
			Comp. All. Ope. Stress	1329.8 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.0 \cdot 428.175) / (1407.13 \cdot 1.0 - 0.6 \cdot 12.0)$$

$$= 3.6704 + 3.1750 = 6.8454 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.048 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 8.825) / (428.175 + 0.6 \cdot 8.825)$$

$$= 29 - 0 = 29 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 12.0) / (425.0 + 0.6 \cdot 12.0)$$

$$= 39 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (12.0 \cdot (428.175 + 0.6 \cdot 8.825)) / (1.0 \cdot 8.825)$$

$$= 589.440 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 1.392 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 6.134, c = 3.175 mm, E* = 1.0



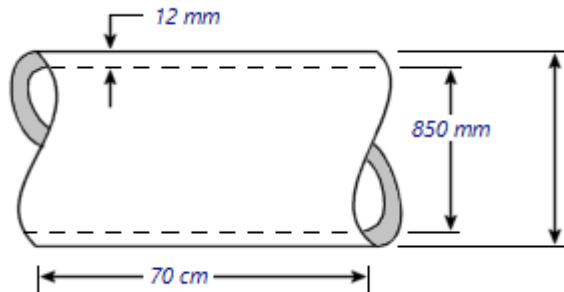
Thickness Ratio = $tr * E^* / (tg - c) = 0.695$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	19.986 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.695	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-40 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: CH SHELL2

Cylindrical Shell Design Information:					
Design Pressure	P	9.317 kgf/cm²	Int. Design Temperature		125 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		70.000 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	850.000 mm	Actual Stress		457.6 kgf/cm²
Spec. Min. Thk t _e , or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		19220.3 cm²	SG of Contents		1.000
Element Volume		397216.1 cm³	Weight of Contents		397.0 kgf
Empty Weight		176.3 kgf	Operating Weight		573.3 kgf
			Comp. All. Ope. Stress		1329.8 kgf/cm²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.317 \cdot 428.175) / (1407.13 \cdot 1.0 - 0.6 \cdot 9.317)$$

$$= 2.8464 + 3.1750 = 6.0214 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.085 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 8.825) / (428.175 + 0.6 \cdot 8.825)$$

$$= 29 - 0.1 = 29 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 12.0) / (425.0 + 0.6 \cdot 12.0)$$

$$= 39 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (9.317 \cdot (428.175 + 0.6 \cdot 8.825)) / (1.0 \cdot 8.825)$$

$$= 457.636 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.392 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 3.936, c = 3.175 mm, E* = 1.0



Thickness Ratio = $tr * E^* / (tg - c) = 0.446$, Temp. Reduction = 41 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	12.865 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.446	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %



Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	25.982 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	15.538 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	44.405 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	21.985 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	17.092 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	17.092 kgf/cm ²
App 27-4	= M.A.W.P.	19.986 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	16.724 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	12.001 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	18.734 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	14.151 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	13.201 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	13.201 kgf/cm ²
App 27-4	= M.A.W.P.	12.865 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
CH. SHELL1	825.6	2386.0	0.346	16.81
SHELL	1280.4	2386.0	0.537	26.07
CH SHELL2	825.6	2386.0	0.346	16.81

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS Inlet Nozzle	Nozzle	1203.20	1203.20	1.000
TS Inlet Nozzle	Pad	1407.13	1407.13	1.000
SS INLET NOZZLE	Nozzle	1203.20	1203.20	1.000
SS INLET NOZZLE	Pad	1407.13	1407.13	1.000
SS OUTLET NOZZL	Nozzle	1203.20	1203.20	1.000
SS OUTLET NOZZL	Pad	1407.13	1407.13	1.000
TS OUT Nozzle	Nozzle	1203.20	1203.20	1.000
TS OUT Nozzle	Pad	1407.13	1407.13	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
CH. COVER	1407.13	1407.13	1.000
CH. FLG1	1407.13	1407.13	1.000
CH. SHELL1	1407.13	1407.13	1.000
SHELL	1407.13	1407.13	1.000
CH SHELL2	1407.13	1407.13	1.000
CH. FLG2	1407.13	1407.13	1.000
CH. COVER2	1407.13	1407.13	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



Tube Material	801.45	801.45	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
TS Inlet Nozzle	369.73	1564.16	0.236
SS INLET NOZZLE	545.37	1564.16	0.349
SS OUTLET NOZZL	545.37	1564.16	0.349
TS OUT Nozzle	369.73	1564.16	0.236

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Test Pressure		25.98 kgf/cm ²
Shell Thickness	ts	12.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	850.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL1

Channel Type:		Cylinder
Channel Test Pressure		16.72 kgf/cm ²
Channel Thickness	tc	12.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	850.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	602
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	7309.00 mm
Straight Tube Length (bet. inner tubsht faces) L		7233.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-214
Tube Material UNS Number		K01807
Is this a Welded Tube		Yes
Tube Material Specification used		Wld. tube

Note:

Welded Tube Material Spec. already includes a Joint Eff. of 0.85. Since ASME checks Long. Tube stress, Jt. Eff. of long. seam is not applicable. Increasing the Tube allowables by dividing by 0.85.

Tube Allowable Stress at Temperature		801.45 kgf/cm ²
Tube Allowable Stress At Ambient		801.45 kgf/cm ²
Tube Allowable Stress at Temper.(for Pressure)		801.45 kgf/cm ²
Tube Allowable Stress at Temper. (for ASME)	St	942.89 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1835.39 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	23.8120 mm
Tube Layout Pattern		Square

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	419.210 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



Length of Expanded Portion of Tube	ltx	38.0000 mm
Distance from Channel Side where Exp. Starts		0.0000 mm
Min. Distance from Shell Side where Exp. Stops		0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	3254.7 cm ²
Perimeter of Tube Layout	Cp	2309.1 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K02700
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	38.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	874.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	727.500 mm
Tube End condition corresponding to Span (l)	k	1.00

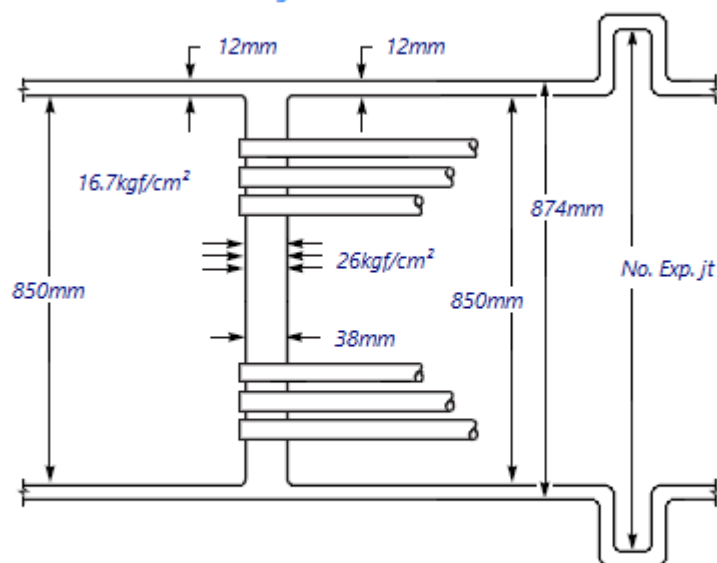
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option	Perform Plastic Calculation	

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Tubesheet Integral w/Shell and Channel



Configuration a:



Messages regarding the Tubesheet & Exchanger Input Data:

*Warning - Length of Expanded portion of Tubes (38.000) is
 = Tubesheet Thickness (38.000 mm).*

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (16.72 \cdot (19.05/2 - 0)) / (801.45 \cdot 1.0 + 0.4 \cdot 16.72)$
 $= 0.1971 + 0.0000 = 0.1971 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	7233.00	9.03	50.0000	0.0134948	977.27

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 153.1568 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5125	19.05	7233.00	37.17	50.0000	0.0007963	724.34

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 724.3384) / (3 \cdot 37.1681) = 25.9842 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:



Total Required Thickness including Corrosion all.	0.5125 mm
Allowable Internal Pressure at Corroded thickness	194.80 kgf/cm ²
Required Internal Design Pressure	16.72 kgf/cm ²
Allowable External Pressure at Corroded thickness	153.16 kgf/cm ²
Required External Design Pressure	25.98 kgf/cm ²
Required Thickness due to Shell Side pressure	0.5125 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \text{ (modified for corrosion if present)} \\ &= 38.0/38.0 \\ &= 1.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(19.05 - 2 \cdot 2.11 (2067025/2067025) \cdot \\ &\quad (942/1407) \cdot (1.0); 19.05 - 2 \cdot 2.11) \\ &= 16.2223 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 419.21 + 19.05 \\ &= 857.470 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (23.812 - 19.05) / 23.812 \\ &= 0.200 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 857.4699/2 \\ &= 428.7350 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 23.812 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 857.47 \cdot 23.812) / (\pi \cdot 857.47^2)) } \\ &= 23.8120 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (23.812 - 16.2223) / 23.812 \\ &= 0.3187 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 425.0/428.735 \\ &= 0.9913 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 425.0/428.735 \\ &= 0.9913 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 602 ((19.05 - 2 \cdot 2.11) / (2 \cdot 428.735))^2 \\ &= 0.8199 \end{aligned}$$

Determine parameter [xs]:



$$= 1 - Nt(dt/(2 \cdot ao))^2$$

$$= 1 - 602 (19.05/(2 \cdot 428.735))^2$$

$$= 0.7029$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 12.0 (850.0 + 12.0)2067025/7233.0$$

$$= 92867.8281 \text{ kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.11 (19.05 - 2.11)2067025/7233.0$$

$$= 320.9021 \text{ kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 92867.828/(602 \cdot 320.902)$$

$$= 0.48072$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kj)$$

$$= 1/(1 + 92867.828/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((850.0 + 12.0)12.0)$$

$$= 0.0179 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.018 \cdot 2067025 \cdot 12.0^3 / (6(1 - 0.3^2))$$

$$= 11692528.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(850.0 + 12.0) \cdot 11692528/38.0^3 \cdot (1 + 38.0 \cdot 0.018 + (38.0 \cdot 0.018)^2/2)$$

$$= 2104821.2500 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= Ds^2/(4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s/(2(Ds + ts)))$$

$$= 850.0^2/(4 \cdot 2067025 \cdot 12.0) \cdot (1 - 850.0 \cdot 0.3/(2(850.0 + 12.0)))$$

$$= 0.62049\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((850.0 + 12.0)12.0)$$

$$= 0.0179 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.018 \cdot 2067025 \cdot 12.0^3 / (6(1 - 0.3^2))$$

$$= 11692528.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta)$$

$$= 6(850.0 + 12.0/\cos(0)) \cdot 11692528/38.0^3 \cdot (1.0 + 38.0 \cdot 0.018 \cdot (1.0 + 1.0)/2 + (38.0 \cdot 0.018)^2/2 \cdot 1.0)$$

$$= 2104821.2 \text{ kgf/cm}^2$$



Channel coefficient [δ_c]:

$$\begin{aligned} &= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c))) \\ &= 850.0^2 / (4 \cdot 2067025 \cdot 12.0) \cdot (1 - 850.0 \cdot 0.3 / (2(850.0 + 12.0))) \\ &= 0.62049E-02 \text{ mm/kgf/cm}^2 \end{aligned}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$\begin{aligned} h/p &= 1.595834 ; \mu^* = 0.318735 ; E^* = 776993 \text{ kgf/cm}^2 \\ E^*/E &= 0.375899 ; \nu^* = 0.315982 \end{aligned}$$

Tube Bundle Stiffness Factor [X_a]:

$$\begin{aligned} &= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)_{ao^2}) / (E^* \cdot L \cdot H^3))^{1/4} \\ &= ((24(1 - 0.316^2)602 \cdot 2067025 \cdot 2.11 \\ &\quad (19.05 - 2.11)428.735^2) / (776993 \cdot 7233.0 \cdot 38.0^3))^{1/4} \\ &= 4.8920 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.013063 ; Z_v = 0.042708 ; Z_m = 0.300797 \\ Z_a &= 0.200760E+02 ; Z_w = 0.042708 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 874.0/857.4699 \\ &= 1.0193 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.32) / 776993 \cdot (2104821 + 2104821 + \\ &\quad 2067025 \cdot \ln(1.02)) \\ &= 3.7407 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*)F \\ &= (1 + 0.316)3.7407 \\ &= 4.9226 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (0.9913 - 1 - 4.9226 \cdot 0.0427) / (1 + 4.9226 \cdot 0.3008) \\ &= -0.088260 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (0.01306 + -0.08826 \cdot 0.04271) / 2 \cdot 4.89204^4 \\ &= 2.6613 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.04271 + -0.08826 \cdot 0.3008) / 2 \cdot 4.89204^4 \\ &= 4.6276 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\psi_s - 1)Z_m)X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.0427 + (0.9913 - 1)0.3008)4.89204^4 / (1 + 4.9226 \cdot 0.3008) \\ &= 9.2553 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$



Calculate parameter [ws]:

$$= ps \cdot ks \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 0.9913 \cdot 11692528 \cdot 0.0179 \cdot 0.006205 (1 + 38.0 \cdot 0.0179)$$

$$= 2158.5215 \text{ mm}^2$$

Calculate parameter [ws*]:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - ws$$

$$= 428.735^2 (0.991^2 - 1) \cdot (0.991 - 1) / 4 - 2158.521$$

$$= -2151.5769 \text{ mm}^2$$

Parameter [wc]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 0.991 \cdot 11692528 [0.006205 (1.0 + 0.018 \cdot 1.0 \cdot 38.0) -$$

$$0 (1.0/38.0 + 1.0 \cdot 0.018/2)]$$

$$= 2158.5215 \text{ mm}^2$$

Parameter [wc*]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - wc$$

$$= 428.73495^2 [(0.99129^2 + 1) \cdot (0.99129 - 1) / 4 - (0.99129 - 1) / 2] - 2158.52148$$

$$= -2151.5769 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs -$$

$$[(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 25.982 \{ 0.703 + 2(1 - 0.703) 0.3 + [2/0.481 \cdot$$

$$(850.0/857.47)^2] 0.3 - [(0.991^2 - 1)/(1.0 \cdot 0.481)] -$$

$$[(1 - 1.0)/(2 \cdot 1.0 \cdot 0.481)] [(0^2 - (850.0)^2)/857.47^2] \}$$

$$= 55.6971 \text{ kgf/cm}^2$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.82 + 2(1 - 0.82) 0.3 + 1/(1.0 \cdot 0.481)] 16.724$$

$$= 50.3092 \text{ kgf/cm}^2$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 602 \cdot 320.902 \cdot 0 / (3.142 \cdot 428.735^2)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 9.255 \cdot 0 / (2 \cdot \pi \cdot 428.735^2)$$

$$= -0.0000 \text{ kgf/cm}^2$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega^*s \cdot Ps - \omega^*c \cdot Pt)$$

$$= -(9.255/428.735^2) \cdot (-2151.577 \cdot 25.982 - -2151.577 \cdot 16.724)$$

$$= 1.0029 \text{ kgf/cm}^2$$

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega^*s \cdot P^*s - \omega^*c \cdot P^*c)$$

$$= (9.255/428.735^2) \cdot (2158.5215 \cdot 0 - 2158.5215 \cdot 0)$$

$$= 0.0000 \text{ kgf/cm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot$$

$$(P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.481 / (1 + 1.0 \cdot 0.481 (2.661 + (0.991 -$$

$$1) 4.628)) \cdot (55.697 - 50.309 + 0 + 0 + 0 + 1.003)$$

$$= 1.3594 \text{ kgf/cm}^2$$



ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega_s \cdot P * s - \omega_c \cdot P * c)) + \\ &\quad W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ &= [((-2151.577 \cdot 25.982 - -2151.577 \cdot 16.724) - \\ &\quad (2158.521 \cdot 0 - 2158.521 \cdot 0)) \cdot 0.01 + \\ &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.92264 \cdot 0.3008) \\ &= -80.2929 \text{ kgf} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} &= Q1 + 2 \cdot Q2 / (P_e \cdot a_o^2) \\ &= -0.088 + 2 \cdot -80.293 / (1.359 \cdot 428.735^2) \\ &= -0.15253 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.076263

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 0.0763 / 0.3187) \cdot (2 \cdot 428.735 / (38.0 - 0))^2 \cdot 1.36 \\ &= 248.4253 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 2651.12 = 3777.85 \text{ kgf/cm}^2, 4.18.233$$

Note: Tubesheet Bending Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\ \text{Tubesheet Bending Stress} &= 1657.1779 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y) \\ &= (2 \cdot 0.2 \cdot 38.0 / 428.735) \cdot 2014.853 \\ &= 71.4269 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 89.1999 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [H_{reqd}] :

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(3.81 ; 3.81) \\ &= 3.8100 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.2896, F_{tmax} = 1.8849$$

First Extreme Tube Axial Stress from among all the tubes [σ_1]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\ &= ((25.98 \cdot 0.7029 - 16.72 \cdot 0.8199) - (1.359) \cdot -0.29) / \\ &\quad (0.8199 - 0.7029) \\ &= 42.2235 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ &= ((25.98 \cdot 0.7029 - 16.72 \cdot 0.8199) - (1.359) \cdot 1.885) / \\ &\quad (0.8199 - 0.7029) \\ &= 16.9722 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 42.224 \text{ kgf/cm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:



$$= \beta T \cdot S_{y,t} = 1743.6229 \text{ kgf/cm}^2$$

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 42.22 \cdot 1.1229 \\ &= 47.41 \text{ kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 1743.623 \\ &= 1957.932 \text{ kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 1743.62 \\ &= 1957.9325 \text{ kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 1743.623 / 1743.623 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 1743.62 \\ &= 1300.8619 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right) \\ &= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right) \\ &= 2.562 \text{ mm} \end{aligned}$$

$$\text{Minimum Required Fillet Weld Leg [afr]: } 1.2810 \text{ mm}$$

$$\text{Minimum Required Groove Weld Leg [agr]: } 1.2810 \text{ mm}$$

Tube-Tubesheet Jt allowable, 1957.93 is $\geq$ tube strength 1957.93 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 2651.12 = 3777.85 \text{ kgf/cm}^2$$

Axial Membrane Stress in Shell [σ_s ,m]:

$$\begin{aligned} &= a_o^2 / (D_s + t_s) t_s \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / (D_s + t_s) t_s \\ &= 428.735^2 / ((850.0 + 12.0) 12.0) \cdot [1.36 + (0.991^2 - 1) \\ &\quad (25.98 - 16.72)] + 425.0^2 \cdot 16.72 / ((850.0 + 12.0) 12.0) \\ &= 313.3382 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 12.0 mm, the minimum Shell length adjacent to the tubesheet should be 181.791 mm

Stress/Force Summary for Loadcase D3 un-corr. ($P_{sd,max}$ + $P_{td,max}$):



Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	248.4	<= 3777.8 kgf/cm ²	Pass
Tubesheet Shear Stress	25.2	<= 2014.9 kgf/cm ²	Pass
Maximum Tube Stress	42.2	<= 1743.6 kgf/cm ²	Pass
Maximum Force on any one Tube	47.4	<= 1957.9 kgf	Pass
Axial Membrane Stress in Shell	313.3	<= 3777.8 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	3.810	38.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
			Bend	Allwd	Shear	Allwd		
D1uc	17.759	...	1361	3778	206	2015	Ps+Pt-Th	D1 Ok
D2uc	20.094	...	1610	3778	231	2015	Ps+Pt-Th	D2 Ok
D3uc	3.810	...	248	3778	25	2015	Ps+Pt-Th	D3 Ok
Max:	20.0944	... mm		0.426		0.115	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

	----	Shell Stresses			----	:	----	Shell Band Stress			----	:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail		
D1uc	100	3777	...	...	:	...	...	...	...	:	Ok		
D2uc	213	3777	...	...	:	...	...	...	...	:	Ok		
D3uc	313	3777	...	...	:	...	...	...	...	:	Ok		
Max RATIO		0.083		...	:		...		...	:			

Tube, Shell and Channel Stress Summary:

Case#	----	Tube Stresses				Tube Loads		Shell Stress		Channel Stress	Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd		Stress Allwd	Stress	Allwd	Fail
D1uc	156	1743	-156	-1743	175	1957	...	3777	...	3777	Ok
D2uc	198	1743	-101	-1743	222	1957	...	3777	...	3777	Ok



D3uc	42	1743	...	...	47	1957	...	3777	...	3777	Ok
Max RATIO	0.113		0.089		0.113		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	20.094	38.000	Ok
Tube thickness :	0.513	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (12.000) adj. to tubesheet: 181.791 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.108 mm
 Due to Pressure + Thermal Shell Compresses by : -0.108 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed Delta Ms UnCorroded	Computed Delta Ms Corroded
1	0.0195	...	...	...	...	...
2	-0.1076	...	...	...	...	...
3	-0.0881	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	51.825	No Calc	No Calc
20	30	No Calc	...	71.825	No Calc	No Calc
30	Tsht	69.997	874	8.825	0.0017272	1023.03
Tsht	40	No Calc	...	8.825	No Calc	No Calc
40	Tsht	723.299	874	8.825	0.00014176	144.513
Tsht	50	No Calc	...	8.825	No Calc	No Calc
50	60	70.0036	874	8.825	0.0017271	1023
60	70	No Calc	...	71.825	No Calc	No Calc
70	80	No Calc	...	51.825	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	55	34.9081	...	No Calc
20	30	75	72.7202	...	No Calc
30	Tsht	12	No Calc	...	13.773
Tsht	40	12	No Calc	...	No Calc
40	Tsht	12	No Calc	...	1.94558
Tsht	50	12	No Calc	...	No Calc
50	60	12	No Calc	...	13.7727
60	70	75	63.3476	...	No Calc
70	80	55	31.6519	...	No Calc

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	69.997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	723.299	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	70.0036	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	443.313	...	443.313	...	44.3313
20	30	176.409	76177.4	167.637	76602.9	17.6409
30	40	176.302	397216	130.133	403173	17.6302
40	50	1821.71	2863308	1344.65	2924861	182.171
50	60	176.302	397216	130.133	403173	17.6302
60	70	176.409	76177.4	167.637	76602.9	17.6409
70	80	443.313	...	443.313	...	44.3313
Total		3413	3810095.50	2826	3884413.50	341



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
30	Liqd	397.037	35	...	...	Liquid: 30
30	Nozl	73.4949	35	-55.2254	...	TS Inlet Nozzle
40	Sadl	111.052	146	54.05	...	Left Saddle
40	Sadl	111.052	585.5	54.05	...	RIGHT SADDLE
40	Liqd	1611.32	361.65	...	...	Liquid: 40
40	Insl	141.12	361.65	...	...	Ins: 30
40	Nozl	252.346	670	66.3912	...	SS INLET NOZZLE
40	Nozl	252.346	55	-66.3912	...	SS OUTLET NOZZL
50	Liqd	397.037	35	...	...	Liquid: 50
50	Nozl	73.4949	35	55.2254	...	TS OUT Nozzle
30	FTsh	138.775	71.9	...	...	Fixed TS
30	Tube	4212.26	435.45	...	...	Tubes
30	RTsh	138.775	799	...	...	Tubesheet
40	Bafl	18.3421	80.4	...	...	Baffles
40	Bafl	18.3421	160.7	...	...	Baffles
40	Bafl	18.3421	241	...	...	Baffles
40	Bafl	18.3421	321.3	...	...	Baffles
40	Bafl	18.3421	401.6	...	...	Baffles
40	Bafl	18.3421	481.9	...	...	Baffles
40	Bafl	18.3421	562.2	...	...	Baffles
40	Bafl	18.3421	642.5	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	222.1
Liquid	2405.4
Insulation	141.1
Nozzles	651.7
Exchanger Components	4636.5
Liquid in Tubes	759.7

Sum of the Detail Weights 8816.5 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	3755.1	3755.1	3755.1	3755.1	3755.1	3755.1
Saddles	222.1	222.1	222.1	222.1	222.1	222.1
Nozzles	651.7	651.7	651.7	651.7	651.7	651.7
Exchanger	4636.5	4636.5	4636.5	4636.5	4636.5	4636.5



Insulation	...	...	141.1	141.1	141.1	141.1
Ope. Liquid	...	...	...	...	...	2405.4
Tube Ope Lqd	...	...	...	...	...	759.7
Test Liquid	...	3808.4	...	...	...	...
Tube Tst Lqd	...	759.7	...	...	...	...
Totals	9265.5	13833.5	9406.6	9406.6	9406.6	12571.7

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	9265.5 kg
Shop Test Wt.	- Fabricated weight + water (full)	13833.5 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	9406.6 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	9406.6 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	9406.6 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	12571.7 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	11926.0 kg

Exchanger Tube Data

Volume of Exchanger tubes :	760024.6 cm ³
Weight of Ope Liq in tubes :	759.7 kg
Weight of Water in tubes :	759.7 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	10414.4	...
20	30	6548.18	3551.57
30	40	19220.3	18692.5
40	50	198600	193147
50	60	19220.3	18692.5
60	70	6548.18	3551.57
70	80	10414.4	...

Total	270965.9	237634.8 cm ²
Total	291.7	255.8 ft ²

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
TS Inlet Nozzl	46.81	52.01	125	300	GR 1.1	0.975	46.806	46.806	46.806
SS INLET NOZZL	46.81	52.01	125	300	GR 1.1	0.591	46.806	46.806	46.806
SS OUTLET NOZZ	46.81	52.01	125	300	GR 1.1	0.015	46.806	46.806	46.806
TS OUT Nozzle	46.81	52.01	125	300	GR 1.1	...	46.806	46.806	46.806

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 46.806 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 52.007 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 46.806 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 52.007 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
TS Inlet Nozzle	302.387	0.50	1	683	12.865	46.806	Pass
SS INLET NOZZLE	551.688	0.50	5	1377	19.986	46.806	Pass
SS OUTLET NOZZL	551.688	0.50	5	0	19.986	46.806	Pass
TS OUT Nozzle	302.387	0.50	0	0	12.865	46.806	Pass

Evaluating Flange on Nozzle [TS Inlet Nozzle]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1) + 4 \cdot \text{abs}(683) \cdot 302.387 \leq \pi \cdot 302.387^3 \cdot [(46.806 - 12.865) + 0.500 \cdot 46.806]$$

$$= 0.84680E+03 \leq 0.49812E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle TS Inlet Nozzle [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(46.806, 0.500 \cdot 46.806 + 46.806 - 846.8 / (\pi \cdot 302.387^3))$$

$$= \min(46.806, 69.235)$$

$$= 46.806 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [SS INLET NOZZLE]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(5) + 4 \cdot \text{abs}(1377) \cdot 551.688 \leq \pi \cdot 551.688^3 \cdot [(46.806 - 19.986) + 0.500 \cdot 46.806]$$

$$= 0.31196E+04 \leq 0.26493E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SS INLET NOZZLE [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(46.806, 0.500 \cdot 46.806 + 46.806 - 3119.6 / (\pi \cdot 551.688^3))$$

$$= \min(46.806, 69.618)$$

$$= 46.806 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [SS OUTLET NOZZL]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(5) + 4 \cdot \text{abs}(0) \cdot 551.688 \leq \pi \cdot 551.688^3 \cdot [(46.806 - 19.986) + 0.500 \cdot 46.806]$$

$$= 0.81548E+02 \leq 0.26493E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SS OUTLET NOZZL [Pmax]:



$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(46.806, 0.500 \cdot 46.806 + 46.806 - 81.5 / (\pi \cdot 551.688^3)) \\ &= \min(46.806, 70.194) \\ &= 46.806 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [TS OUT Nozzle]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(0) + 4 \cdot \text{abs}(0) 302.387 \leq \pi \cdot 302.387^3 [(46.806 - 12.865) + .500 \cdot 46.806] \\ &= 0.00000E+00 \leq 0.49812E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle TS OUT Nozzle [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(46.806, 0.500 \cdot 46.806 + 46.806 - 0.0 / (\pi \cdot 302.387^3)) \\ &= \min(46.806, 70.210) \\ &= 46.806 \text{ kgf/cm}^2 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 9.672

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 0.99 • 1.0 • 0.6 • 57.5²)

= 1966.1 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.1 • 0.893 • 1966.125

= 118.3 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	0.98	1.0000	1.0000	1.0000	57.50	0.8934
20	0.98	1.0000	1.0000	1.0000	57.50	0.8934
30	0.98	1.0000	1.0000	1.0000	57.50	0.8934
40	0.98	1.0000	1.0000	1.0000	57.50	0.8934
50	0.98	1.0000	1.0000	1.0000	57.50	0.8934
60	0.98	1.0000	1.0000	1.0000	57.50	0.8934
70	0.98	1.0000	1.0000	1.0000	57.50	0.8934

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
10	20	98	122.4	673.2	200.571	13.265
20	30	98	102	1275	200.571	25.1232
30	40	98	104.88	7341.6	200.571	144.662
40	50	98	115.68	83671.4	200.571	1648.7
50	60	98	104.88	7341.6	200.571	144.662
60	70	98	102	1275	200.571	25.1232
70	80	98	122.4	673.2	200.571	13.265

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.0939 • 1.0) • 1.0
= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 1300.44 • 0.0939 • 1.0
= 122.1 kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.0939 • 11704 • 1.0
= 732.3 kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------

From	To	Height cm	Weight kgf	Ope Load kgf
10	20	51	1300.44	122.054
20	30	42.5	1300.44	122.054
30	40	42.5	1300.44	122.054
40	Sad1	42.5	1300.44	122.054
Sad1	50	42.5	1300.44	122.054
40	50	42.5	1300.44	122.054
50	60	42.5	1300.44	122.054
60	70	42.5	1300.44	122.054
70	80	51	1300.44	122.054

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	458.000 cm
Center of Gravity of the Liquid	453.900 cm
Center of Gravity of the Insulation	453.900 cm
Center of Gravity of the Nozzles	454.558 cm
Center of Gravity of the Tubesheet(s)	453.900 cm
Center of Gravity of the Tubes	453.900 cm
Center of Gravity of the Baffles	453.700 cm
Center of Gravity of Bare Shell New and Cold	453.982 cm
Center of Gravity of Bare Shell Corroded	453.997 cm
Vessel CG in the Operating Condition	454.035 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	454.072 cm
Vessel CG in the Test Condition	454.018 cm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

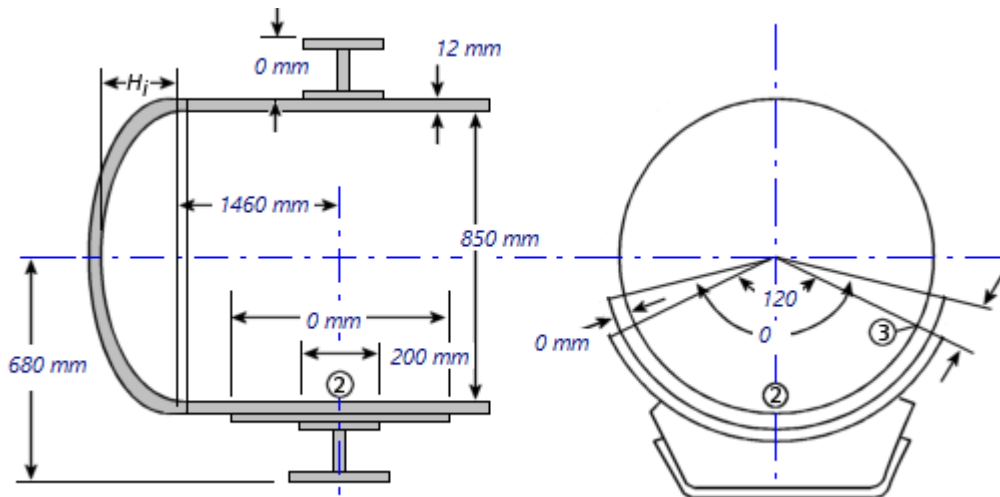
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{432.5876 \cdot 8.825}, 2 \cdot 1460.0)$$

$$= 296.3871 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	432.59	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	723.30	cm
Distance from Saddle to Vessel tangent	a1 or a	1460.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	680.00	mm
Coefficient of Friction	mu	0.50	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		8994.66	kgf
Pressure used in Analysis	P	11.976	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	230.91	1407.13
Long. Stress at Bottom of Midspan	356.15	1407.13
Long. Stress at Top of Saddles	751.74	1407.13
Long. Stress at Bottom of Saddles	39.56	1407.13
Tangential Shear in Shell	164.48	1125.71
Circ. Stress at Horn of Saddle	1001.57	1758.92
Circ. Compressive Stress in Shell	26.14	1407.13



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(2014.8/2 + 0) \cdot 680.0/770.0$
 $= 2669.0 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(190.61, 0) \cdot 680.0/4395.0005$
 $= 29.5 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(1098.48, 0) \cdot 680.0/4395.0005$
 $= 170.0 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(1098/2 + 0) \cdot 680.0/770.0$
 $= 1455.1 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 6326 + \max(29, 2669, 170, 1455)$
 $= 8994.7 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1966.921(0.0 + \pi/4(0.874 \cdot 1.2)^2)$
 $= 1869.210 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	9105.71	kgf
Transverse Shear Load	1007.40	kgf
Longitudinal Shear Load	1098.48	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 770.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -8995 \cdot 146.0 [1 - (1 - 146.0/723.3 + (43.259^2 - 0.0^2)/(2 \cdot 146.0 \cdot 723.3))/(1 + (4 \cdot 0.0)/(3 \cdot 723.3))]$
 $= -2534.4 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 8995 \cdot 723/4((1 + 2(43^2 - 0^2)/(723^2))/(1 + (4 \cdot 0)/(3 \cdot 723)) - 4 \cdot 146/723)$
 $= 3248.8 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 11.976 \cdot 432.588/(2 \cdot 8.825) - 3248.8/(\pi \cdot 432.6^2 \cdot 8.825)$
 $= 230.91 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$



$$= 11.976 \cdot 432.588 / (2 \cdot 8.825) + 3248.8 / (\pi \cdot 432.6^2 \cdot 8.825)$$

$$= 356.15 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 11.976 \cdot 432.588 / (2 \cdot 8.825) - 2534.4 / (0.1066 \cdot \pi \cdot 432.6^2 \cdot 8.825)$$

$$= 751.74 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 11.976 \cdot 432.588 / (2 \cdot 8.825) + 2534.4 / (0.1923 \cdot \pi \cdot 432.6^2 \cdot 8.825)$$

$$= 39.56 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 8995(723.3 - 2 \cdot 146.0) / (723.3 + (4 \cdot 0.0 / 3))$$

$$= 5363.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 5363.47 / (432.5876 \cdot 8.825)$$

$$= 164.48 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(432.588 \cdot 8.825)}$$

$$= 48.194 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 8995 \cdot 0.1 / (8.825(200.0 + 48.19 + 48.19))$$

$$= -26.14 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -8995 / (4 \cdot 8.825(200.0 + 48.194 + 48.194)) - 3 \cdot 0.0529 \cdot 8995 / (2 \cdot 8.825^2)$$

$$= -1001.57 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(432.588 \cdot 8.825)}, 2 \cdot 146.0)$$

$$= 296.39 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 4395.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 4395.0(125.0 - 21.1)$$

$$= 5.600 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	770.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		125.0	°C
Saddle Yield Stress	Sy	2396.2	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(428.175 \cdot 8.825)} + 225.0)8.825^3) / 12$$



$$= 1.838 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1.838 + 28.319(94.895 - 4.412)^2$$

$$= 2320.354 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	320.9	8.8	4.4	28.3	12495.7	0.232E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.173E+04
Web	16.0	215.0	128.3	34.4	441438.0	0.171E+04
BasePlate	150.0	16.0	243.8	24.0	585179.9	0.533E+04
Totals	...	...	...	113.7	1079141.1	0.111E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1079141/113.719$$

$$= 94.895 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 8994.66$$

$$= 1830.6084 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 21.4357 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1437.7354 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 680.0 - 428.18 \cdot \sin(120.0/2) / 1.0472$$

$$= 325.902 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 596.5999 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 51.0386 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1597.4836 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(8995 + 111)150.0 / (2 \cdot 770.0 \cdot 1597.484)}$$

$$= 12.906 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 777.6908 / (3 - 1)$$

$$= 388.845 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 388.8454 \cdot 150.0/2$$

$$= 291.634 \text{ cm}^2$$



Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 9105.712 / (770.0 \cdot 150.0)$
 $= 7.884 \text{ kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 291.6 \cdot 7.88$
 $= 2299.165 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 29.44 + 31.108$
 $= 60.548 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 2299.2 / 60.5476$
 $= 37.973 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.161E+04
Web	194.4	16.0	8.0	31.1	24886.1	742.
Totals	...	...	...	60.5	342838.0	0.235E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 200.0 - 16.0$
 $= 184.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 342838.031 / 60.548$
 $= 56.623 \text{ mm}$

Distance to Centroid [C1]:
 $= \max(y_{tot}, \text{Saddle Width} - y_{tot})$
 $= \max(56.623, 200.0 - 56.623)$
 $= 143.377 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{ (2349.8 / 60.548)}$
 $= 62.297 \text{ mm}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 680.0 - \cos(120.0 / 2) (425.0 + 12.0 + 12.0) - 16.0$
 $= 439.500 \text{ mm}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120.0 / 2) (425.0 + 12.0 + 12.0)$
 $= 777.691 \text{ mm}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 770.0 - 50.8$
 $= 719.200 \text{ mm}$

Length of Outer Rib, step 4, [Dd]:
 $= (L_w - B_d) / 2$
 $= (777.691 - 719.2) / 2$
 $= 29.245 \text{ mm}$



Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{439.5^2 + 29.245^2}$$

$$= 440.472 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 2004815 / 2396.2}$$

$$= 128.510$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 440.472 / 62.297$$

$$= 7.071$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1098.5 / (2 \cdot 770.0) \cdot 388.845 \cdot 440.47 / 2$$

$$= 61.086 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.0706 < 128.5104$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.07)^2 / (2 \cdot 128.51^2)) 2396 /$$

$$(5/3 + 3 \cdot (7.07) / (8 \cdot 128.51) - (7.07^3) / (8 \cdot 128.51^3))$$

$$= 1418 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 37.97 / 1418.02 + (61.09 \cdot 143.377 / 23497792) / 1597.48$$

$$= 0.050$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.219E+04
Web	388.8	16.0	8.0	62.2	49772.2	655.
Totals	...	...	...	91.7	367724.2	0.284E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 367724.156 / 91.655$$

$$= 40.120 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(40.12, \text{abs}(200.0 - 40.12))$$

$$= 159.880 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 680.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 215.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2842.2 / 91.655}$$

$$= 55.687 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 215.0 / 55.687$$

$$= 3.861$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1098.483 / (2 \cdot 770.0)$$

$$= 0.713 \text{ kgf/mm}$$



Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.713 \cdot 388.845 \cdot 215.0$$

$$= 59.633 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 59.633 / 177774.297$$

$$= 33.544 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (3.8609 < 128.5104) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.86)^2 / (2 \cdot 128.51^2)) 2396 /$$

$$(5/3 + 3 \cdot (3.86) / (8 \cdot 128.51) - (3.86^3) / (8 \cdot 128.51^3))$$

$$= 1427 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 57.77 / 1427.44 + (59.63 \cdot 159.88 / 2842.247) / 1597.48$$

$$= 0.061$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.2586 cm ²
Saddle Load Q0 (Weight)	Q0	6436.8 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	170.0 kgf
Maximum Transverse Force	Ft	1007.4 kgf
Maximum Longitudinal Force	Fl	1098.5 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1098.5 / (2.259 \cdot 2)$$

$$= 243.2 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1007.4 / (2.259 \cdot 2)$$

$$= 223.0 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarear!]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1098.48 / (1101.26 \cdot 2.0)$$

$$= 0.4987 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 680.0 \cdot 1007.4 + 0.0$$

$$= 685.03 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$



= 685.03/6436.76
 = 106.43 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.4987, 0.0]
 = 0.4987 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, tr = 6.134, c = 3.175 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.695, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = min(b + $1.56 \cdot \sqrt{Rm \cdot t}$, 2a)
 = min(200.0 + $1.56 \cdot \sqrt{432.5876 \cdot 8.825}$, 2 * 1460.0)
 = 296.3871 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	432.59	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	723.30	cm
Distance from Saddle to Vessel tangent	a1 or a	1460.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	680.00	mm
Coefficient of Friction	mu	0.50	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		8779.49	kgf
Pressure used in Analysis	P	11.976	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	232.41	1407.13
Long. Stress at Bottom of Midspan	354.65	1407.13
Long. Stress at Top of Saddles	740.78	1407.13
Long. Stress at Bottom of Saddles	45.64	1407.13
Tangential Shear in Shell	160.54	1125.71
Circ. Stress at Horn of Saddle	977.61	1758.92
Circ. Compressive Stress in Shell	25.52	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(2014.8/2 + 0) \cdot 680.0/770.0$$

$$= 2669.0 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind; Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(190.61, 0) \cdot 680.0/4395.0005$$

$$= 29.5 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(1098.48, 0) \cdot 680.0/4395.0005$$

$$= 170.0 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1098/2 + 0) \cdot 680.0/770.0$$

$$= 1455.1 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 6111 + \max(29, 2669, 170, 1455)$$

$$= 8779.5 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1966.921(0.0 + \pi/4(0.874 \cdot 1.2)^2)$$

$$= 1869.210 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8890.54	kgf
Transverse Shear Load	1007.40	kgf
Longitudinal Shear Load	1098.48	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 770.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -8779 \cdot 146.0 [1 - (1 - 146.0/723.3 + (43.259^2 - 0.0^2)/(2 \cdot 146.0 \cdot 723.3))/(1 + (4 \cdot 0.0)/(3 \cdot 723.3))]$$

$$= -2473.8 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm)/(3L)) - 4a/L)$$

$$= 8779 \cdot 723/4((1 + 2(43^2 - 0^2)/(723^2))/(1 + (4 \cdot 0)/(3 \cdot 723)) - 4 \cdot 146/723)$$

$$= 3171.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 11.976 \cdot 432.588/(2 \cdot 8.825) - 3171.0/(\pi \cdot 432.6^2 \cdot 8.825)$$

$$= 232.41 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 11.976 \cdot 432.588/(2 \cdot 8.825) + 3171.0/(\pi \cdot 432.6^2 \cdot 8.825)$$

$$= 354.65 \text{ kgf/cm}^2$$



Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 11.976 \cdot 432.588 / (2 \cdot 8.825) - 2473.8 / (0.1066 \cdot \pi \cdot 432.6^2 \cdot 8.825)$
 $= 740.78 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 11.976 \cdot 432.588 / (2 \cdot 8.825) + 2473.8 / (0.1923 \cdot \pi \cdot 432.6^2 \cdot 8.825)$
 $= 45.64 \text{ kgf/cm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a) / (L + (4 \cdot h_m / 3))$
 $= 8779(723.3 - 2 \cdot 146.0) / (723.3 + (4 \cdot 0.0 / 3))$
 $= 5235.2 \text{ kgf}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 5235.17 / (432.5876 \cdot 8.825)$
 $= 160.54 \text{ kgf/cm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(432.588 \cdot 8.825)}$
 $= 48.194 \text{ mm}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$
 $= -0.7603 \cdot 8779 \cdot 0.1 / (8.825(200.0 + 48.19 + 48.19))$
 $= -25.52 \text{ kgf/cm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$
 $= -8779 / (4 \cdot 8.825(200.0 + 48.194 + 48.194)) - 3 \cdot 0.0529 \cdot 8779 / (2 \cdot 8.825^2)$
 $= -977.61 \text{ kgf/cm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(200.0 + 1.56 \cdot \sqrt{(432.588 \cdot 8.825)}, 2 \cdot 146.0)$
 $= 296.39 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	770.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		125.0	°C
Saddle Yield Stress	Sy	2396.2	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(428.175 \cdot 8.825)} + 225.0)^3) / 12$
 $= 1.838 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 1.838 + 28.319(94.895 - 4.412)^2$
 $= 2320.354 \text{ cm}^4$

B	D	Y	A	AY	I + AD ⁽²⁾
mm	mm	mm	cm ²	mm ³	cm ⁴



Shell	320.9	8.8	4.4	28.3	12495.7	0.232E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.173E+04
Web	16.0	215.0	128.3	34.4	441438.0	0.171E+04
BasePlate	150.0	16.0	243.8	24.0	585179.9	0.533E+04
Totals	...	...	...	113.7	1079141.1	0.111E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1079141/113.719$$

$$= 94.895 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 8779.493$$

$$= 1786.8171 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 20.9230 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1437.7354 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 680.0 - 428.18 \cdot \sin(120.0/2) / 1.0472$$

$$= 325.902 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 582.3281 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 49.8177 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1597.4836 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}) }$$

$$= \sqrt{ 3(8779 + 111) 150.0 / (2 \cdot 770.0 \cdot 1597.484) }$$

$$= 12.752 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 777.6908 / (3 - 1)$$

$$= 388.845 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388.8454 \cdot 150.0/2$$

$$= 291.634 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 8890.545 / (770.0 \cdot 150.0)$$

$$= 7.697 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 291.6 \cdot 7.7$$

$$= 2244.836 \text{ kgf}$$



Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 29.44 + 31.108$$

$$= 60.548 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 2244.8/60.5476$$

$$= 37.076 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.161E+04
Web	194.4	16.0	8.0	31.1	24886.1	742.
Totals	...	...	...	60.5	342838.0	0.235E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 16.0$$

$$= 184.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 342838.031/60.548$$

$$= 56.623 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(56.623, 200.0 - 56.623)$$

$$= 143.377 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2349.8/60.548)}$$

$$= 62.297 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 680.0 - \cos(120.0/2)(425.0 + 12.0 + 12.0) - 16.0$$

$$= 439.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2)(425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 770.0 - 50.8$$

$$= 719.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd)/2$$

$$= (777.691 - 719.2)/2$$

$$= 29.245 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(439.5^2 + 29.245^2)}$$

$$= 440.472 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2004815/2396.2)}$$

$$= 128.510$$



Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 440.472/62.297$$

$$= 7.071$$

Bending Moment $[Rm]$:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 1098.5 / (2 \cdot 770.0) \cdot 388.845 \cdot 440.47/2$$

$$= 61.086 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ (7.0706 < 128.5104) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.07)^2 / (2 \cdot 128.51^2)) 2396 /$$

$$(5/3 + 3 \cdot (7.07) / (8 \cdot 128.51) - (7.07^3) / (8 \cdot 128.51^3))$$

$$= 1418 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 37.08/1418.02 + (61.09 \cdot 143.377/23497792) / 1597.48$$

$$= 0.049$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.219E+04
Web	388.8	16.0	8.0	62.2	49772.2	655.
Totals	...	...	...	91.7	367724.2	0.284E+04

Distance to Centroid from Datum $[ytot]$:

$$= AY / A$$

$$= 367724.156/91.655$$

$$= 40.120 \text{ mm}$$

Distance to Centroid $[C1]$:

$$= \max(ytot, \text{abs}(Saddle Width - ytot))$$

$$= \max(40.12, \text{abs}(200.0 - 40.12))$$

$$= 159.880 \text{ mm}$$

Length of Inner Rib $[L]$:

$$= Saddle Height - Outside Radius - Bpthk$$

$$= 680.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 215.000 \text{ mm}$$

Radius of Gyration $[r]$:

$$= \sqrt{ (Total Inertia / Total Area) }$$

$$= \sqrt{ (2842.2/91.655) }$$

$$= 55.687 \text{ mm}$$

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 215.0/55.687$$

$$= 3.861$$

Unit Force $[Force,u]$:

$$= FI / (2 \cdot Baseplate Length)$$

$$= 1098.483 / (2 \cdot 770.0)$$

$$= 0.713 \text{ kgf/mm}$$

Moment at base of inner Rib $[Mbase,c]$:

$$= Unit Force \cdot e \cdot L$$

$$= 0.713 \cdot 388.845 \cdot 215.0$$

$$= 59.633 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma B,base,c]$:

$$= Bending Moment / Section Modulus$$

$$= 59.633/177774.297$$

$$= 33.544 \text{ kgf/cm}^2$$



Compressive Allowable, $KL/r < C_c$ (3.8609 < 128.5104) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)^4 / (8 \cdot C_c^4) - (KL/r)^6 / (8 \cdot C_c^6))$$

$$= (1 - (3.86)^2 / (2 \cdot 128.51^2)) 2396 / (5/3 + 3 \cdot (3.86)^4 / (8 \cdot 128.51^4) - (3.86)^6 / (8 \cdot 128.51^6))$$

$$= 1427 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 56.41 / 1427.44 + (59.63 \cdot 159.88 / 2842.247) / 1597.48$$

$$= 0.061$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BlArea	2.2586 cm ²
Saddle Load QO (Weight)	QO	6221.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	170.0 kgf
Maximum Transverse Force	Ft	1007.4 kgf
Maximum Longitudinal Force	Fl	1098.5 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1098.5 / (2.259 \cdot 2)$$

$$= 243.2 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1007.4 / (2.259 \cdot 2)$$

$$= 223.0 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1098.48 / (1101.26 \cdot 2.0)$$

$$= 0.4987 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 680.0 \cdot 1007.4 + 0.0$$

$$= 685.03 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 685.03 / 6221.59$$

$$= 110.11 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.4987, 0.0]$$

$$= 0.4987 \text{ cm}^2$$



MDMT Calculations for: RIGHT SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, tr = 6.134, c = 3.175 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.695$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

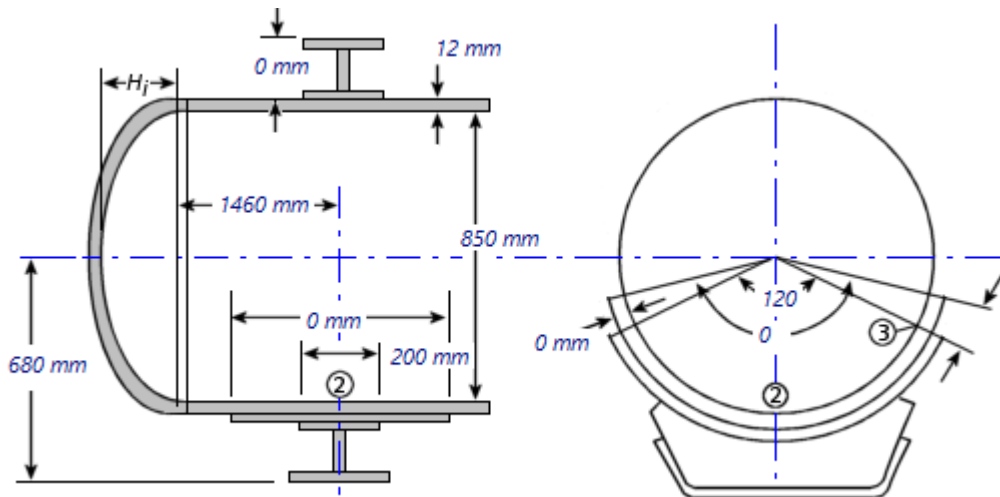
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{432.5876 \cdot 8.825}, 2 \cdot 1460.0)$$

$$= 296.3871 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	432.59	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	723.30	cm
Distance from Saddle to Vessel tangent	a1 or a	1460.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	680.00	mm
Coefficient of Friction	mu	0.50	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		7883.63	kgf
Pressure used in Analysis	P	26.025	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	582.97	2386.01
Long. Stress at Bottom of Midspan	692.74	2386.01
Long. Stress at Top of Saddles	1039.46	2386.01
Long. Stress at Bottom of Saddles	415.26	2386.01
Tangential Shear in Shell	144.16	1413.05
Circ. Stress at Horn of Saddle	877.86	3579.01
Circ. Compressive Stress in Shell	22.91	2386.01



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (664.9/2 + 0) \cdot 680.0/770.0$
 $= 880.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(62.9, 0) \cdot 680.0/4395.0005$
 $= 9.7 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 7003 + \max(10, 881, 0, 0)$
 $= 7883.6 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1966.921 (0.0 + \pi/4 (0.874 \cdot 1.2)^2)$
 $= 1869.210 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7994.69	kgf
Transverse Shear Load	332.44	kgf
Longitudinal Shear Load	62.90	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 770.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -7884 \cdot 146.0 [1 - (1 - 146.0/723.3 + (43.259^2 - 0.0^2)/(2 \cdot 146.0 \cdot 723.3))/(1 + (4 \cdot 0.0)/(3 \cdot 723.3))]$
 $= -2221.4 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 7884 \cdot 723/4 ((1 + 2(43^2 - 0^2)/(723^2))/(1 + (4 \cdot 0)/(3 \cdot 723)) - 4 \cdot 146/723)$
 $= 2847.5 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 $= 26.025 \cdot 432.588/(2 \cdot 8.825) - 2847.5/(\pi \cdot 432.6^2 \cdot 8.825)$
 $= 582.97 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 $= 26.025 \cdot 432.588/(2 \cdot 8.825) + 2847.5/(\pi \cdot 432.6^2 \cdot 8.825)$
 $= 692.74 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$
 $= 26.025 \cdot 432.588/(2 \cdot 8.825) - 2221.4/(0.1066 \cdot \pi \cdot 432.6^2 \cdot 8.825)$
 $= 1039.46 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$



$$= 26.025 \cdot 432.588 / (2 \cdot 8.825) + 2221.4 / (0.1923 \cdot \pi \cdot 432.6^2 \cdot 8.825)$$

$$= 415.26 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 7884(723.3 - 2 \cdot 146.0)/(723.3 + (4 \cdot 0.0/3))$$

$$= 4701.0 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4700.97 / (432.5876 \cdot 8.825)$$

$$= 144.16 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(432.588 \cdot 8.825)}$$

$$= 48.194 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 7884 \cdot 0.1 / (8.825(200.0 + 48.19 + 48.19))$$

$$= -22.91 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q/(2 \cdot t^2)$$

$$= -7884/(4 \cdot 8.825(200.0 + 48.194 + 48.194)) -$$

$$3 \cdot 0.0529 \cdot 7884/(2 \cdot 8.825^2)$$

$$= -877.86 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(432.588 \cdot 8.825)}, 2 \cdot 146.0)$$

$$= 296.39 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	770.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3)/12$$

$$= ((1.56 \cdot \sqrt{(428.175 \cdot 8.825)} + 225.0)^3)/12$$

$$= 1.838 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 1.838 + 28.319(94.895 - 4.412)^2$$

$$= 2320.354 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	320.9	8.8	4.4	28.3	12495.7	0.232E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.173E+04
Web	16.0	215.0	128.3	34.4	441438.0	0.171E+04
BasePlate	150.0	16.0	243.8	24.0	585179.9	0.533E+04
Totals	...	...	...	113.7	1079141.1	0.111E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1079141/113.719$$



$$= 94.895 \text{ mm}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 7883.633 \\ &= 1604.4902 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 18.7880 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 680.0 - 428.18 \cdot \sin(120.0/2) / 1.0472 \\ &= 325.902 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 522.9073 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 44.7343 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\ &= \sqrt{(3(7884 + 111) \cdot 150.0) / (2 \cdot 770.0 \cdot 1767.415))} \\ &= 11.497 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0) \\ &= 777.691 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 777.6908 / (3 - 1) \\ &= 388.845 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 388.8454 \cdot 150.0/2 \\ &= 291.634 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 7994.685 / (770.0 \cdot 150.0) \\ &= 6.922 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 291.6 \cdot 6.92 \\ &= 2018.634 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 29.44 + 31.108 \\ &= 60.548 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P/Ar \\ &= 2018.6/60.5476 \end{aligned}$$



= 33.340 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.161E+04
Web	194.4	16.0	8.0	31.1	24886.1	742.
Totals	...	...	...	60.5	342838.0	0.235E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 200.0 - 16.0
 = 184.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 342838.031/60.548
 = 56.623 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(56.623, 200.0 - 56.623)
 = 143.377 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{2349.8/60.548}$
 = 62.297 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 680.0 - cos(120.0/2)(425.0 + 12.0 + 12.0) - 16.0
 = 439.500 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(120.0/2)(425.0 + 12.0 + 12.0)
 = 777.691 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 770.0 - 50.8
 = 719.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (777.691 - 719.2)/2
 = 29.245 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$
 = $\sqrt{439.5^2 + 29.245^2}$
 = 440.472 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2004815/2651.1}$
 = 122.176

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 440.472/62.297
 = 7.071

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 62.9 / (2 • 770.0) • 388.845 • 440.47/2



= 3.498 kgf-m.

Compressive Allowable, $KL/r < Cc$ (7.0706 < 122.1764) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.07)^2 / (2 \cdot 122.18^2)) 2651 / (5/3 + 3 \cdot (7.07) / (8 \cdot 122.18) - (7.07^3) / (8 \cdot 122.18^3))$$

$$= 1568 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 33.34/1567.62 + (3.5 \cdot 143.377/23497792) / 1767.42$$

$$= 0.022$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.219E+04
Web	388.8	16.0	8.0	62.2	49772.2	655.
Totals	...	...	...	91.7	367724.2	0.284E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 367724.156/91.655$$

$$= 40.120 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(40.12, \text{abs}(200.0 - 40.12))$$

$$= 159.880 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 680.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 215.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2842.2/91.655) }$$

$$= 55.687 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 215.0/55.687$$

$$= 3.861$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 62.9 / (2 \cdot 770.0)$$

$$= 0.041 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.041 \cdot 388.845 \cdot 215.0$$

$$= 3.415 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3.415/177774.297$$

$$= 1.921 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (3.8609 < 122.1764) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (3.86)^2 / (2 \cdot 122.18^2)) 2651 / (5/3 + 3 \cdot (3.86) / (8 \cdot 122.18) - (3.86^3) / (8 \cdot 122.18^3))$$

$$= 1579 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$



$$= 50.73/1578.66 + (3.41 \cdot 159.88/2842.247)/1767.42$$

$$= 0.033$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.2586 cm ²
Saddle Load QO (Weight)	QO	7113.9 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	9.7 kgf
Maximum Transverse Force	Ft	332.4 kgf
Maximum Longitudinal Force	Fl	62.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 62.9 / (2.259 \cdot 2)$$

$$= 13.9 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 332.4 / (2.259 \cdot 2)$$

$$= 73.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 62.9 / (1052.29 \cdot 2.0)$$

$$= 0.0299 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 680.0 \cdot 332.44 + 0.0$$

$$= 226.06 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 226.06/7113.93$$

$$= 31.78 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0299, 0.0]$$

$$= 0.0299 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$



$$= \min(200.0 + 1.56 \sqrt{(432.5876 \cdot 8.825)}, 2 \cdot 1460.0)$$

$$= 296.3871 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	432.59	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	723.30	cm
Distance from Saddle to Vessel tangent	a1 or a	1460.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	680.00	mm
Coefficient of Friction	mu	0.50	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		7630.42	kgf
Pressure used in Analysis	P	26.025	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	584.73	2386.01
Long. Stress at Bottom of Midspan	690.97	2386.01
Long. Stress at Top of Saddles	1026.56	2386.01
Long. Stress at Bottom of Saddles	422.41	2386.01
Tangential Shear in Shell	139.53	1413.05
Circ. Stress at Horn of Saddle	849.66	3579.01
Circ. Compressive Stress in Shell	22.18	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(664.9/2 + 0) \cdot 680.0/770.0$
 $= 880.8 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(62.9, 0) \cdot 680.0/4395.0005$
 $= 9.7 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 6750 + \max(10, 881, 0, 0)$
 $= 7630.4 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1966.921(0.0 + \pi/4(0.874 \cdot 1.2)^2)$
 $= 1869.210 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7741.47	kgf
Transverse Shear Load	332.44	kgf
Longitudinal Shear Load	62.90	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \text{Baseplate Length}$



= 770.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -7630 \cdot 146.0 [1 - (1 - 146.0/723.3 + (43.259^2 - 0.0^2) / (2 \cdot 146.0 \cdot 723.3)) / (1 + (4 \cdot 0.0) / (3 \cdot 723.3))]$$

$$= -2150.0 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 7630 \cdot 723 / 4 ((1 + 2(43^2 - 0^2)/(723^2)) / (1 + (4 \cdot 0) / (3 \cdot 723)) - 4 \cdot 146 / 723)$$

$$= 2756.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 26.025 \cdot 432.588 / (2 \cdot 8.825) - 2756.0 / (\pi \cdot 432.6^2 \cdot 8.825)$$

$$= 584.73 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 26.025 \cdot 432.588 / (2 \cdot 8.825) + 2756.0 / (\pi \cdot 432.6^2 \cdot 8.825)$$

$$= 690.97 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 26.025 \cdot 432.588 / (2 \cdot 8.825) - 2150.0 / (0.1066 \cdot \pi \cdot 432.6^2 \cdot 8.825)$$

$$= 1026.56 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 26.025 \cdot 432.588 / (2 \cdot 8.825) + 2150.0 / (0.1923 \cdot \pi \cdot 432.6^2 \cdot 8.825)$$

$$= 422.41 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 7630(723.3 - 2 \cdot 146.0) / (723.3 + (4 \cdot 0.0 / 3))$$

$$= 4550.0 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 4549.98 / (432.5876 \cdot 8.825)$$

$$= 139.53 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(432.588 \cdot 8.825)}$$

$$= 48.194 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 7630 \cdot 0.1 / (8.825(200.0 + 48.19 + 48.19))$$

$$= -22.18 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -7630 / (4 \cdot 8.825(200.0 + 48.194 + 48.194)) - 3 \cdot 0.0529 \cdot 7630 / (2 \cdot 8.825^2)$$

$$= -849.66 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(432.588 \cdot 8.825)}, 2 \cdot 146.0)$$



= 296.39 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	770.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(428.175 \cdot 8.825)} + 225.0)^3) / 12$$

$$= 1.838 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 1.838 + 28.319 (94.895 - 4.412)^2$$

$$= 2320.354 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	320.9	8.8	4.4	28.3	12495.7	0.232E+04
Wearplate	225.0	12.0	14.8	27.0	40027.5	0.173E+04
Web	16.0	215.0	128.3	34.4	441438.0	0.171E+04
BasePlate	150.0	16.0	243.8	24.0	585179.9	0.533E+04
Totals	...	...	...	113.7	1079141.1	0.111E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1079141 / 113.719$$

$$= 94.895 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 7630.417$$

$$= 1552.9553 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 18.1845 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 680.0 - 428.18 \cdot \sin(120.0/2) / 1.0472$$

$$= 325.902 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 506.1120 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 43.2975 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:



$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(7630 + 111)150.0 / (2 \cdot 770.0 \cdot 1767.415)}$$

$$= 11.313 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 777.6908 / (3 - 1)$$

$$= 388.845 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388.8454 \cdot 150.0 / 2$$

$$= 291.634 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 7741.469 / (770.0 \cdot 150.0)$$

$$= 6.703 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 291.6 \cdot 6.7$$

$$= 1954.698 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 29.44 + 31.108$$

$$= 60.548 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1954.7 / 60.5476$$

$$= 32.284 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.161E+04
Web	194.4	16.0	8.0	31.1	24886.1	742.
Totals	...	...	...	60.5	342838.0	0.235E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 16.0$$

$$= 184.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 342838.031 / 60.548$$

$$= 56.623 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(56.623, 200.0 - 56.623)$$

$$= 143.377 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2349.8 / 60.548}$$

$$= 62.297 \text{ mm}$$



Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 680.0 - \cos(120.0/2)(425.0 + 12.0 + 12.0) - 16.0$$

$$= 439.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2)(425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 770.0 - 50.8$$

$$= 719.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (777.691 - 719.2) / 2$$

$$= 29.245 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{\text{Lv}^2 + \text{Dd}^2}$$

$$= \sqrt{439.5^2 + 29.245^2}$$

$$= 440.472 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2004815 / 2651.1)}$$

$$= 122.176$$

Slenderness ratio [KL/r]:

$$= \text{KL} / r$$

$$= 1 \cdot 440.472 / 62.297$$

$$= 7.071$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 62.9 / (2 \cdot 770.0) \cdot 388.845 \cdot 440.47 / 2$$

$$= 3.498 \text{ kgf-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($7.0706 < 122.1764$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (7.07)^2 / (2 \cdot 122.18^2)) 2651 /$$

$$(5/3 + 3 \cdot (7.07) / (8 \cdot 122.18) - (7.07^3) / (8 \cdot 122.18^3))$$

$$= 1568 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc} / \text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 32.28 / 1567.62 + (3.5 \cdot 143.377 / 23497792) / 1767.42$$

$$= 0.022$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	184.0	108.0	29.4	317951.9	0.219E+04
Web	388.8	16.0	8.0	62.2	49772.2	655.
Totals	...	...	...	91.7	367724.2	0.284E+04

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 367724.156 / 91.655$$

$$= 40.120 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{abs}(\text{Saddle Width} - \text{ytot}))$$

$$= \max(40.12, \text{abs}(200.0 - 40.12))$$

$$= 159.880 \text{ mm}$$



Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 680.0 - \cos(449.0/2)(16.0 + 0.0 + 0.0) - 0.0$$

$$= 215.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2842.2/91.655}$$

$$= 55.687 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 215.0/55.687$$

$$= 3.861$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 62.9/(2 \cdot 770.0)$$

$$= 0.041 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.041 \cdot 388.845 \cdot 215.0$$

$$= 3.415 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3.415/177774.297$$

$$= 1.921 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.8609 < 122.1764$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.86)^2 / (2 \cdot 122.18^2)) 2651 /$$

$$(5/3 + 3 \cdot (3.86) / (8 \cdot 122.18) - (3.86^3) / (8 \cdot 122.18^3))$$

$$= 1579 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 49.12/1578.66 + (3.41 \cdot 159.88/2842.247) / 1767.42$$

$$= 0.032$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BlArea	2.2586 cm ²
Saddle Load Q0 (Weight)	Q0	6860.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	9.7 kgf
Maximum Transverse Force	Ft	332.4 kgf
Maximum Longitudinal Force	Fl	62.9 kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 62.9/(2.259 \cdot 2)$$

$$= 13.9 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 332.4/(2.259 \cdot 2)$$

$$= 73.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$



Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 62.9/(1052.29 • 2.0)
= 0.0299 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 680.0 • 332.44 + 0.0
= 226.06 kgf-m.

Eccentricity (e):
= Rmom / QO
= 226.06/6860.71
= 32.95 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0299, 0.0]
= 0.0299 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: TS Inlet Nozzle

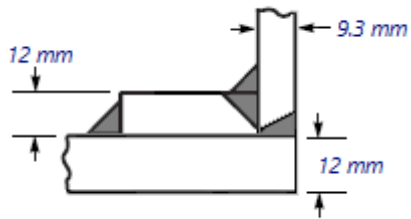
From: 30

Pressure for Reinforcement Calculations	P	9.317	kgf/cm ²
Temperature for Internal Pressure	Temp	125	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		53.45	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1120	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	373.0500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS Inlet Nozzle

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	10.020 in.
Actual Thickness Used in Calculation	0.365 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.32 \cdot 428.175 / (1407 \cdot 1.0 - 0.6 \cdot 9.32)$
 $= 2.8463 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.32 \cdot 130.43 / (1203 \cdot 1.0 - 0.6 \cdot 9.32)$
 $= 1.0147 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	521.7160 mm
Parallel to Vessel Wall, opening length	d	260.8580 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0625 mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

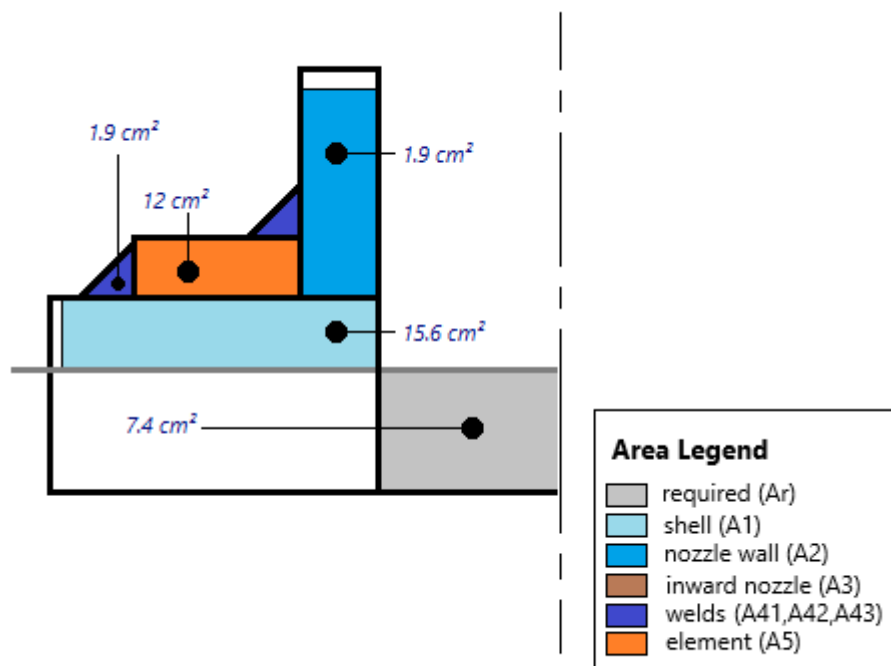


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7.425	NA	NA
Area in Shell	A1	15.596	NA	NA
Area in Nozzle Wall	A2	1.917	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	31.368	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (260.858 \cdot 2.8463 \cdot 1.0 + 2 \cdot 6.096 \cdot 2.8463 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 7.425 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 260.858 (1.0 \cdot 8.825 - 1.0 \cdot 2.846) - 2 \cdot 6.096 \\
 &\quad (1.0 \cdot 8.825 - 1.0 \cdot 2.846) \cdot (1 - 1.0) \\
 &= 15.596 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 22.06) \cdot (6.1 - 1.01) \cdot 0.8551 \\
 &= 1.917 \text{ cm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0
 \end{aligned}$$



= 1.855 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
= (373.05 - 273.05) * 12.0 * 1.0
= 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 4.1897 mm
Wall Thickness per UG16(b), tr16b = 4.6750 mm
Wall Thickness, shell/head, internal pressure trb1 = 6.0213 mm
Wall Thickness tb1 = max(trb1, tr16b) = 6.0213 mm
Wall Thickness tb2 = max(trb2, tr16b) = 4.6750 mm
Wall Thickness per table UG-45 tb3 = 11.2776 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[11.278, max(6.0213, 4.675)]
= 6.0213 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(4.1897, 6.0213)
= 6.0213 mm

Available Nozzle Neck Thickness = 0.875 * 9.271 = 8.112 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	84.4,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2923.6 kgf/cm ²	Passed
Occasional	:	97.4,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	57.9,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 8.112, tr = 1.015, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.206, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 8.112, tr = 1.015, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.206, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 8.112, tr = 1.015, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.206, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 2.846, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.323, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 8.112, tr = 1.015, c = 3.175 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.206, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 9.32/52.01 = 0.179

Weld Size Calculations, Description: TS Inlet Nozzle

Intermediate Calc. for nozzle/shell Welds Tmin 6.0960 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2672 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.4125 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = (Ar - A1) • Sv
 = (7.4247 - 15.596) • 1407
 = 0.00 kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
 = (1.9172 + 12.0 + 1.8551 - 0.0 • 0.86) • 1407
 = 22193.67 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 = (π/2) • Dlo • Wo • 0.49 • Snw
 = (3.1416/2.0) • 273.05 • 10.0 • 0.49 • 1203
 = 25287. kgf

Shear, Pad Element Weld [Spew]:
 = (π/2) • DP • WP • 0.49 • SEW
 = (3.1416/2.0) • 373.05 • 10.0 • 0.49 • 1407
 = 40403. kgf

Tension, Pad Groove Weld [Tpgw]:
 = (π/2) • Dlo • Wgpn • 0.74 • Seg
 = (3.1416/2) • 273.05 • 12.0 • 0.74 • 1407
 = 53593. kgf

Shear, Nozzle Groove Weld [Sngw]:
 = (π/2) • (Dlr+Wgnva) • (Wgnva-Can) • 0.60 • Sng
 = (3.14/2)•(260.858+7.112)•(7.112-3.175)•0.6•1407
 = 13991. kgf



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNGW) = (40403 + 13991) = 54395 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 54394 kgf, must exceed $W = 0 \text{ kgf}$ or $W1 = 22193 \text{ kgf}$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 16 kgf/cm^2

The Drop for this Nozzle is : 22.5252 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.5252 mm

Input Echo, WRC297 Item 1, Description: TS Inlet Nozzl:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	850.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	254.508	mm
Nozzle Thickness	9.2710	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	373.050	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	9.32	kgf/cm ²
Radial Load	P	-683.24	kgf
Circumferential Shear	Vc	1019.77	kgf
Longitudinal Shear	Vl	1019.77	kgf
Circumferential Moment	Mc	0.76	kgf-m.
Longitudinal Moment	Ml	1.02	kgf-m.
Torsional Moment	Mt	1.27	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(438.588(12.0 - 3.175))}$



= 478.355 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 877.175 mm
 Nozzle Outside Diameter (d) = 273.050 mm
 Vessel Thickness used (T) = 20.825 mm
 Nozzle Thickness used (t) = 6.096 mm
 T / t = 3.416
 d / t = 44.792

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.020$$

Nr/P = 0.048
 Mr/P = 0.059
 M0/P = 0.057
 N0/P = 0.201
 MrD/Mc = 0.132
 NrDL/Mc = 0.086
 M0d/Mc = 0.161
 N0DL/Mc = 0.296
 MrD/Ml = 0.078
 NrDL/Ml = 0.056
 M0D/Ml = 0.086
 N0DL/Ml = 0.408

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	31	31	31	31
Outplane Bending (P)	53	-53	53	-53
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	186	196	186	196
Outplane Stress Summary	271	174	273	174

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	93	93	93	93
Inplane Stress Summary	156	45	156	44

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top	Al Top	Bu Bottom	Bl Bottom
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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	272	175	274	174

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	31	31	31	31
Outplane Bending (P)	53	-53	53	-53
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	93	93	93	93
Outplane Stress Summary	178	71	179	70

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	186	196	186	196
Inplane Stress Summary	249	148	250	147

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-11	-11	11	11

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	251	150	252	149

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	31	31	31	31
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	233	244	233	244
Hoop Stress Summary	265	276	265	276

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	389	-389	389	-389
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	116	116	116	116
Axial Stress Summary	516	-256	522	-261

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	38	38	-38	-38
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	39	39	-38	-38

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	521	538	528	543

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	31	31	31	31
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	233	244	233	244
Hoop Stress Summary	265	276	265	276

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	13	13	13	13
Axial Bending (P)	389	-389	389	-389
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	116	116	116	116
Axial Stress Summary	515	-255	523	-262

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-38	-38	38	38
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-38	-38	39	39

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	520	537	529	545

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(432.588(12.0 - 3.175))}$
 $= 476.946 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 865.175 mm
 Nozzle Outside Diameter (d) = 373.050 mm
 Vessel Thickness used (T) = 8.825 mm
 Nozzle Thickness used (t) = 56.096 mm
 $T / t = 0.157$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.269$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.048
 Mr/P = 0.078
 M0/P = 0.024
 N0/P = 0.043
 MrD/Mc = 0.251
 NrDL/Mc = 0.093
 M0d/Mc = 0.075
 N0DL/Mc = 0.073
 MrD/Ml = 0.062
 NrDL/Ml = 0.049
 M0D/Ml = 0.019
 N0DL/Ml = 0.044

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	37	37	37	37
Outplane Bending (P)	124	-124	124	-124
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	447	456	447	456
Outplane Stress Summary	608	370	609	370

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	41	41	41	41
Inplane Bending (P)	408	-408	408	-408
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-1	1	1	-1
Inplane Pressure Stress	223	223	223	223
Inplane Stress Summary	672	-142	675	-144

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			19	19	-19	-19
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			19	19	-19	-19

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			678	514	680	516

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			37	37	37	37
Outplane Bending (P)			124	-124	124	-124
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			-1	1	1	-1
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			223	223	223	223
Outplane Stress Summary			384	138	387	136

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			41	41	41	41
Inplane Bending (P)			408	-408	408	-408
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			-3	3	3	-3
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			447	456	447	456
Inplane Stress Summary			893	93	901	86

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-19	-19	19	19
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-19	-19	19	19

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			894	145	902	143

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	186	196	186	196	186	196	186	196
Circ. Pl (SUS)	31	31	31	31	7	7	7	7
Circ. Q (SUS)	53	-53	53	-53	55	-55	55	-55
Long. Pm (SUS)	93	93	93	93	93	93	93	93
Long. Pl (SUS)	7	7	7	7	31	31	31	31
Long. Q (SUS)	55	-55	55	-55	53	-53	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	186.0	196.0	186.0	196.0	186.0	196.0	186.0	196.0
Pm+Pl (SUS)	218.0	227.9	218.0	227.9	194.7	204.5	194.7	204.5
Pm+Pl+Q (Total)	271.0	174.9	271.0	174.9	249.7	149.5	249.7	149.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	196.00	1407.13	Passed
Pm+Pl (SUS)	227.95	2110.70	Passed
Pm+Pl+Q (TOTAL)	271.04	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	447	456	447	456	447	456	447	456
Circ. Pl (SUS)	37	37	37	37	41	41	41	41
Circ. Q (SUS)	124	-124	124	-124	405	-405	411	-411
Long. Pm (SUS)	223	223	223	223	223	223	223	223
Long. Pl (SUS)	41	41	41	41	37	37	37	37
Long. Q (SUS)	407	-407	409	-409	123	-123	125	-125
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	19	19	-19	-19	-19	-19	19	19
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	447.0	456.0	447.0	456.0	447.0	456.0	447.0	456.0
Pm+Pl (SUS)	485.6	494.6	485.6	494.6	489.6	498.5	489.6	498.5
Pm+Pl+Q (Total)	676.3	513.4	678.1	515.4	893.7	143.9	899.7	141.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	456.00	1407.13	Passed
Pm+Pl (SUS)	498.51	2110.70	Passed
Pm+Pl+Q (TOTAL)	899.70	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	233	244	233	244	233	244	233	244
Circ. Pl (SUS)	31	31	31	31	31	31	31	31
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	116	116	116	116	116	116	116	116
Long. Pl (SUS)	13	13	13	13	13	13	13	13
Long. Q (SUS)	387	-387	391	-391	385	-385	393	-393
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	38	38	-38	-38	-38	-38	38	38
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	233.0	244.0	233.0	244.0	233.0	244.0	233.0	244.0
Pm+Pl (SUS)	274.0	284.3	274.0	284.3	274.0	284.3	274.0	284.3
Pm+Pl+Q (Total)	521.6	538.4	525.5	542.4	519.6	536.4	527.5	544.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	244.00	1203.20	Passed
Pm+Pl (SUS)	284.30	1804.80	Passed
Pm+Pl+Q (TOTAL)	544.33	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SS INLET NOZZLE

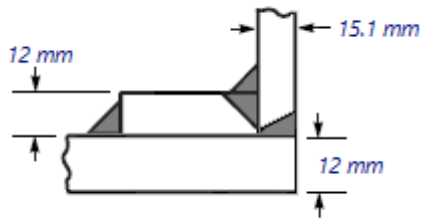
From: 40

Pressure for Reinforcement Calculations	P	11.952	kgf/cm ²
Temperature for Internal Pressure	Temp	125	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		762.25	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	608.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SS INLET NOZZLE

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	18.812 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 11.95 \cdot 428.175 / (1407 \cdot 1.0 - 0.6 \cdot 11.95)$
 $= 3.6556 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 11.95 \cdot 242.09 / (1203 \cdot 1.0 - 0.6 \cdot 11.95)$
 $= 2.4193 \text{ mm}$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	968.3496 mm
Parallel to Vessel Wall, opening length	d	484.1748 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0625 mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

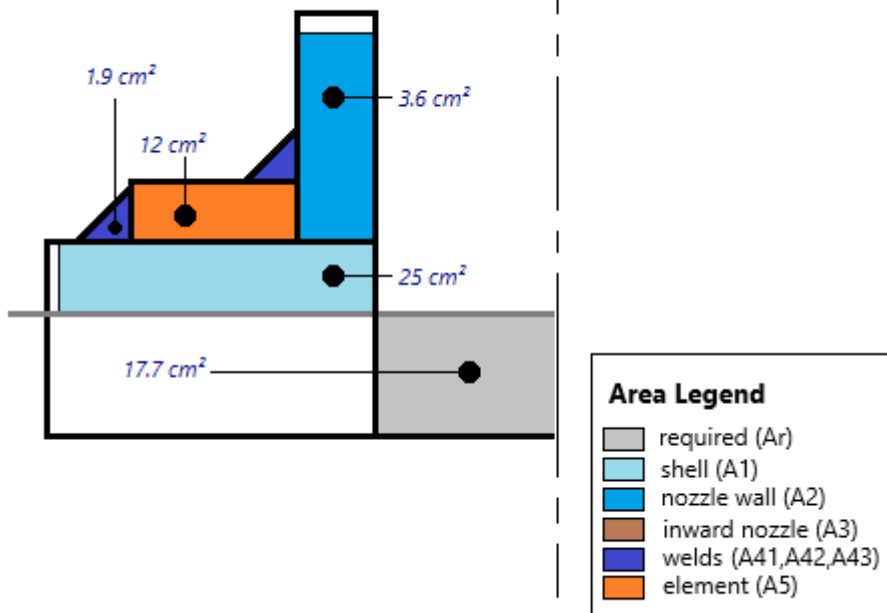


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	17.700	NA	NA
Area in Shell	A1	25.029	NA	NA
Area in Nozzle Wall	A2	3.582	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	42.466	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (484.1748 \cdot 3.6556 \cdot 1.0 + 2 \cdot 11.9126 \cdot 3.6556 \cdot 1.0 \cdot (1 - 1.0))$
 $= 17.700 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 484.175(1.0 \cdot 8.825 - 1.0 \cdot 3.656) - 2 \cdot 11.913$
 $(1.0 \cdot 8.825 - 1.0 \cdot 3.6556) \cdot (1 - 1.0)$
 $= 25.029 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 22.06) \cdot (11.91 - 2.42) \cdot 0.8551$
 $= 3.582 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$



= 1.855 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
= (608.0 - 508.0) * 12.0 * 1.0
= 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 5.5943 mm
Wall Thickness per UG16(b), tr16b = 4.6750 mm
Wall Thickness, shell/head, internal pressure trb1 = 6.8306 mm
Wall Thickness tb1 = max(trb1, tr16b) = 6.8306 mm
Wall Thickness tb2 = max(trb2, tr16b) = 4.6750 mm
Wall Thickness per table UG-45 tb3 = 11.5062 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[11.506, max(6.8306, 4.675)]
= 6.8306 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(5.5943, 6.8306)
= 6.8306 mm

Available Nozzle Neck Thickness = 0.875 * 15.088 = 13.202 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	111.3,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2896.7 kgf/cm ²	Passed
Occasional	:	118.5,	Allowable	:	1600.3 kgf/cm ²	Passed
Shear	:	3.3,	Allowable	:	842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.202, tr = 2.419, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.241, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.414, Temp. Reduction = 47 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.414, Temp. Reduction = 47 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 12.0, tr = 3.656, c = 3.175 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.414$, Temp. Reduction = 47 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 3.656, c = 3.175 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.414$, Temp. Reduction = 47 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $11.95/52.01 = 0.230$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 726.2622 mm
 Parallel to Vessel Wall, opening length d 363.1311 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 22.0625 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		11.800	11.800
Area Available in Shell A1		12.514	12.514
Area Available in Nozzle Wall A2		3.582	3.582
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.855	1.855
Area Available in Pad A5		0.000	12.000
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		16.951	29.951

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SS INLET NOZZLE

Intermediate Calc. for nozzle/shell Welds Tmin 11.9126 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm
 Pad Weld 4.4125 = $0.5 * T_{minPad}$ 7.0700 = $0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = $(A_r - A_1) * S_v$
 = $(17.6995 - 25.0289) * 1407$
 = 0.00 kgf



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (3.5818 + 12.0 + 1.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 24536.04 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 47045. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 608.0 \cdot 10.0 \cdot 0.49 \cdot 1407$
 $= 65850. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 508.0 \cdot 12.0 \cdot 0.74 \cdot 1407$
 $= 99708. \text{ kgf}$

Shear, Nozzle Groove Weld [Sngw]:
 $= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$
 $= (3.14/2) \cdot (484.175+12.0) \cdot (12.0-3.175) \cdot 0.6 \cdot 1407$
 $= 58070. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNGW) = (65850 + 58070) = 123920 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 123920 kgf, must exceed W = 0 kgf or W1 = 24536 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 20 kgf/cm²

The Drop for this Nozzle is : 84.2523 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.2523 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SS INLET NOZZL:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	850.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	477.825	mm
Nozzle Thickness	15.0876	mm



Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	608.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	11.95	kgf/cm ²
Radial Load	P	-1376.68	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	3.06	kgf-m.
Longitudinal Moment	Ml	4.08	kgf-m.
Torsional Moment	Mt	5.10	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.581) is > 0.5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(438.588(12.0 - 3.175))}$
 $= 713.305 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	877.175	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	20.825	mm
Nozzle Thickness used	(t) =	11.913	mm
T / t	=	1.748	
d / t	=	42.644	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.759$$

Nr/P	=	0.050
Mr/P	=	0.074
M0/P	=	0.036
N0/P	=	0.145
MrD/Mc	=	0.180
NrDL/Mc	=	0.085
M0d/Mc	=	0.099
N0DL/Mc	=	0.184
MrD/Ml	=	0.077
NrDL/Ml	=	0.042
M0D/Ml	=	0.037
N0DL/Ml	=	0.224

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	45	45	45	45
Outplane Bending (P)	68	-68	68	-68
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	239	251	239	251
Outplane Stress Summary	353	229	354	229

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	15	15	15	15
Inplane Bending (P)	140	-140	140	-140
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	119	119	119	119
Inplane Stress Summary	275	-4	277	-5

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	353	233	354	235

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	45	45	45	45
Outplane Bending (P)	68	-68	68	-68
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	119	119	119	119
Outplane Stress Summary	233	98	235	96

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	15	15	15	15
Inplane Bending (P)	140	-140	140	-140
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	239	251	239	251
Inplane Stress Summary	394	128	397	125

Vessel Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	394	128	397	125

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	45	45	45	45
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	284	298	284	298
Hoop Stress Summary	330	344	330	345

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	7	7	7	7
Axial Bending (P)	284	-284	284	-284
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	142	142	142	142
Axial Stress Summary	431	-132	435	-136

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	431	477	435	481

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	45	45	45	45
Hoop Bending (P)	0	0	0	0



Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	284	298	284	298

Hoop Stress Summary	330	344	330	344
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	7	7	7	7
Axial Bending (P)	284	-284	284	-284
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	142	142	142	142

Axial Stress Summary	430	-131	437	-137
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-2	-2	2	2
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	430	475	437	482
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(432.588(12.0 - 3.175))}$$

$$= 711.896 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	865.175 mm
Nozzle Outside Diameter	(d) =	608.000 mm
Vessel Thickness used	(T) =	8.825 mm
Nozzle Thickness used	(t) =	61.913 mm
T / t	=	0.143
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.958$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.043
Mr/P	=	0.069
M0/P	=	0.021
N0/P	=	0.037
MrD/Mc	=	0.242
NrDL/Mc	=	0.088
M0d/Mc	=	0.069
N0DL/Mc	=	0.065
MrD/Ml	=	0.043
NrDL/Ml	=	0.037



MØD/M1 = 0.014
 NØDL/M1 = 0.032

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	65	65	65	65
Outplane Bending (P)	218	-218	218	-218
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	573	585	573	585
Outplane Stress Summary	857	433	859	432

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	75	75	75	75
Inplane Bending (P)	734	-734	734	-734
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-2	2	2	-2
Inplane Pressure Stress	286	286	286	286
Inplane Stress Summary	1095	-369	1100	-373

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1095	803	1100	806

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	65	65	65	65
Outplane Bending (P)	218	-218	218	-218
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	286	286	286	286
Outplane Stress Summary	568	136	574	131

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	75	75	75	75
Inplane Bending (P)	734	-734	734	-734
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	573	585	573	585
Inplane Stress Summary	1374	-64	1394	-81

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-2	-2	2	2

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	1374	200	1394	213

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	239	251	239	251	239	251	239	251
Circ. Pl (SUS)	45	45	45	45	15	15	15	15
Circ. Q (SUS)	68	-68	68	-68	139	-139	141	-141
Long. Pm (SUS)	119	119	119	119	119	119	119	119
Long. Pl (SUS)	15	15	15	15	45	45	45	45
Long. Q (SUS)	140	-140	140	-140	68	-68	68	-68
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	239.0	251.0	239.0	251.0	239.0	251.0	239.0	251.0
Pm+Pl (SUS)	284.0	296.0	284.0	296.0	254.0	266.0	254.0	266.0
Pm+Pl+Q (Total)	352.0	234.0	352.0	234.0	393.0	127.0	395.0	125.0

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	251.00	1407.13	Passed
Pm+Pl (SUS)	296.01	2110.70	Passed
Pm+Pl+Q (TOTAL)	395.01	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	573	585	573	585	573	585	573	585
Circ. Pl (SUS)	65	65	65	65	75	75	75	75
Circ. Q (SUS)	218	-218	218	-218	725	-725	743	-743
Long. Pm (SUS)	286	286	286	286	286	286	286	286
Long. Pl (SUS)	75	75	75	75	65	65	65	65
Long. Q (SUS)	732	-732	736	-736	216	-216	220	-220
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	573.0	585.0	573.0	585.0	573.0	585.0	573.0	585.0
Pm+Pl (SUS)	638.0	650.0	638.0	650.0	648.0	660.0	648.0	660.0
Pm+Pl+Q (Total)	1093	803	1097	807	1373	200	1391	214

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	585.00	1407.13	Passed
Pm+Pl (SUS)	660.01	2110.70	Passed
Pm+Pl+Q (TOTAL)	1391.00	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	284	298	284	298	284	298	284	298
Circ. Pl (SUS)	45	45	45	45	45	45	45	45
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	142	142	142	142	142	142	142	142
Long. Pl (SUS)	7	7	7	7	7	7	7	7
Long. Q (SUS)	283	-283	285	-285	281	-281	287	-287
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	284.0	298.0	284.0	298.0	284.0	298.0	284.0	298.0
Pm+Pl (SUS)	329.0	343.0	329.0	343.0	329.0	343.0	329.0	343.0
Pm+Pl+Q (Total)	432.0	477.0	434.0	479.0	430.0	475.0	436.0	481.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	298.00	1203.20	Passed
Pm+Pl (SUS)	343.02	1804.80	Passed
Pm+Pl+Q (TOTAL)	481.02	3609.61	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: SS OUTLET NOZZL

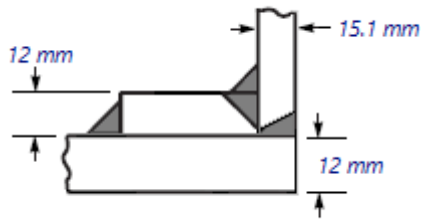
From: 40

Pressure for Reinforcement Calculations	P	12.000	kgf/cm ²
Temperature for Internal Pressure	Temp	125	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		147.25	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	608.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SS OUTLET NOZZL

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	18.812 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.0 \cdot 428.175 / (1407 \cdot 1.0 - 0.6 \cdot 12.0)$
 $= 3.6703$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.0 \cdot 242.09 / (1203 \cdot 1.0 - 0.6 \cdot 12.0)$
 $= 2.4290$ mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	968.3496 mm
Parallel to Vessel Wall, opening length	d	484.1748 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0625 mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

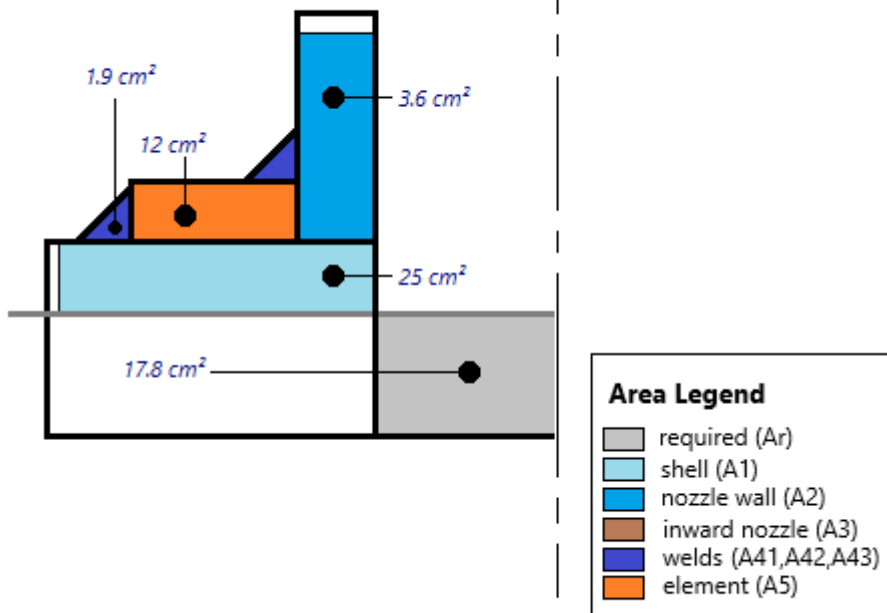


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	17.771	NA	NA
Area in Shell	A1	24.958	NA	NA
Area in Nozzle Wall	A2	3.578	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	42.391	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (484.1748 \cdot 3.6703 \cdot 1.0 + 2 \cdot 11.9126 \cdot 3.6703 \cdot 1.0 \cdot (1 - 1.0))$
 $= 17.771 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 484.175(1.0 \cdot 8.825 - 1.0 \cdot 3.67) - 2 \cdot 11.913$
 $(1.0 \cdot 8.825 - 1.0 \cdot 3.6703) \cdot (1 - 1.0)$
 $= 24.958 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 22.06) \cdot (11.91 - 2.43) \cdot 0.8551$
 $= 3.578 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$



= 1.855 cm²

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_{r4} \\ &= (608.0 - 508.0) \cdot 12.0 \cdot 1.0 \\ &= 12.000 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.6040 mm
Wall Thickness per UG16(b),	tr16b = 4.6750 mm
Wall Thickness, shell/head, internal pressure	trb1 = 6.8453 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 6.8453 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.6750 mm
Wall Thickness per table UG-45	tb3 = 11.5062 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.506, \max(6.8453, 4.675)] \\ &= 6.8453 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(5.604, 6.8453) \\ &= 6.8453 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 0.875 * 15.088 = 13.202 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 126.6,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2881.4 kgf/cm ²	Passed
Occasional	: 119.0,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 32.0,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.202, tr = 2.429, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.242, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.0, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.416, Temp. Reduction = 47 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.0, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.416, Temp. Reduction = 47 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 12.0, tr = 3.67, c = 3.175 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.416$, Temp. Reduction = 47 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 3.67, c = 3.175 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.416$, Temp. Reduction = 47 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $12.00 / 52.01 = 0.231$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 726.2622 mm
 Parallel to Vessel Wall, opening length d 363.1311 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 22.0625 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		11.848	11.848
Area Available in Shell A1		12.479	12.479
Area Available in Nozzle Wall A2		3.578	3.578
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.855	1.855
Area Available in Pad A5		0.000	12.000
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		16.912	29.912

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SS OUTLET NOZZL

Intermediate Calc. for nozzle/shell Welds Tmin 11.9126 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * W_o$ mm
 Pad Weld 4.4125 = $0.5 * T_{minPad}$ 7.0700 = $0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = $(A_r - A_1) * S_v$
 = $(17.7707 - 24.9577) * 1407$
 = 0.00 kgf



Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (3.5782 + 12.0 + 1.8551 - 0.0 \cdot 0.86) \cdot 1407$$

$$= 24530.87 \text{ kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 1203$$

$$= 47045. \text{ kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 608.0 \cdot 10.0 \cdot 0.49 \cdot 1407$$

$$= 65850. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 508.0 \cdot 12.0 \cdot 0.74 \cdot 1407$$

$$= 99708. \text{ kgf}$$

Shear, Nozzle Groove Weld [Sngw]:

$$= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$$

$$= (3.14/2) \cdot (484.175+12.0) \cdot (12.0-3.175) \cdot 0.6 \cdot 1407$$

$$= 58070. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNGW) = (65850 + 58070) = 123920 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 123920 kgf, must exceed W = 0 kgf or W1 = 24530 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 20 kgf/cm²

The Drop for this Nozzle is : 84.2523 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.2523 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SS OUTLET NOZZ:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	850.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	477.825	mm
Nozzle Thickness	15.0876	mm



Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	608.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	12.00	kgf/cm ²
Radial Load	P	1376.68	kgf
Circumferential Shear	Vc	2039.53	kgf
Longitudinal Shear	Vl	2039.53	kgf
Circumferential Moment	Mc	3.06	kgf-m.
Longitudinal Moment	Ml	4.08	kgf-m.
Torsional Moment	Mt	5.10	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.581) is > 0.5
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(438.588(12.0 - 3.175))}$
 $= 713.305 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	877.175	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	20.825	mm
Nozzle Thickness used	(t) =	11.913	mm
T / t	=	1.748	
d / t	=	42.644	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.759$$

Nr/P	=	0.050
Mr/P	=	0.074
M0/P	=	0.036
N0/P	=	0.145
MrD/Mc	=	0.180
NrDL/Mc	=	0.085
M0d/Mc	=	0.099
N0DL/Mc	=	0.184
MrD/Ml	=	0.077
NrDL/Ml	=	0.042
M0D/Ml	=	0.037
N0DL/Ml	=	0.224

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-45	-45	-45	-45
Outplane Bending (P)	-68	68	-68	68
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	240	252	240	252
Outplane Stress Summary	125	275	127	275

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-15	-15	-15	-15
Inplane Bending (P)	-140	140	-140	140
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	120	120	120	120
Inplane Stress Summary	-36	246	-35	244

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	12	12	-12	-12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	164	279	164	279

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-45	-45	-45	-45
Outplane Bending (P)	-68	68	-68	68
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	120	120	120	120
Outplane Stress Summary	5	143	7	142

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	-15	-15	-15	-15
Inplane Bending (P)	-140	140	-140	140
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	240	252	240	252
Inplane Stress Summary	82	379	85	376

Vessel Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-12	-12	12	12

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	84	379	87	376

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-45	-45	-45	-45
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	285	299	285	299
Hoop Stress Summary	239	253	240	254

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-7	-7	-7	-7
Axial Bending (P)	-284	284	-284	284
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	142	142	142	142
Axial Stress Summary	-150	421	-146	417

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	21	21	-21	-21
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	21	21	-21	-21

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	392	423	389	420

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-45	-45	-45	-45
Hoop Bending (P)	0	0	0	0



Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	285	299	285	299
Hoop Stress Summary	239	253	239	254

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-7	-7	-7	-7
Axial Bending (P)	-284	284	-284	284
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	142	142	142	142
Axial Stress Summary	-152	422	-145	416

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-21	-21	21	21
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-21	-21	21	21

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	394	425	387	418

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(432.588(12.0 - 3.175))}$$

$$= 711.896 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	865.175 mm
Nozzle Outside Diameter	(d) =	608.000 mm
Vessel Thickness used	(T) =	8.825 mm
Nozzle Thickness used	(t) =	61.913 mm
T / t	=	0.143
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.958$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.043
Mr/P	=	0.069
M0/P	=	0.021
N0/P	=	0.037
MrD/Mc	=	0.242
NrDL/Mc	=	0.088
M0d/Mc	=	0.069
N0DL/Mc	=	0.065
MrD/Ml	=	0.043
NrDL/Ml	=	0.037



MØD/M1 = 0.014
 NØDL/M1 = 0.032

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	-65	-65	-65	-65
Outplane Bending (P)	-218	218	-218	218
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	576	588	576	588
Outplane Stress Summary	290	741	292	740

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	-75	-75	-75	-75
Inplane Bending (P)	-734	734	-734	734
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-2	2	2	-2
Inplane Pressure Stress	288	288	288	288
Inplane Stress Summary	-525	948	-520	945

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	24	24	-24	-24
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	24	24	-24	-24

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	817	951	814	947

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	-65	-65	-65	-65
Outplane Bending (P)	-218	218	-218	218
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-2	2	2	-2
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	288	288	288	288
Outplane Stress Summary	0	443	6	438

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	-75	-75	-75	-75
Inplane Bending (P)	-734	734	-734	734
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-9	9	9	-9
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	576	588	576	588
Inplane Stress Summary	-244	1255	-224	1238

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-24	-24	24	24
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-24	-24	24	24

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	249	1256	235	1239

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	240	252	240	252	240	252	240	252
Circ. Pl (SUS)	-45	-45	-45	-45	-15	-15	-15	-15
Circ. Q (SUS)	-68	68	-68	68	-141	141	-139	139
Long. Pm (SUS)	120	120	120	120	120	120	120	120
Long. Pl (SUS)	-15	-15	-15	-15	-45	-45	-45	-45
Long. Q (SUS)	-140	140	-140	140	-68	68	-68	68
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	240.0	252.0	240.0	252.0	240.0	252.0	240.0	252.0
Pm+Pl (SUS)	196.6	208.4	196.6	208.4	226.0	237.9	226.0	237.9
Pm+Pl+Q (Total)	163.8	279.2	163.8	279.2	85.8	378.6	87.8	376.6

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	252.00	1407.13	Passed
Pm+Pl (SUS)	237.88	2110.70	Passed
Pm+Pl+Q (TOTAL)	378.61	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	576	588	576	588	576	588	576	588
Circ. Pl (SUS)	-65	-65	-65	-65	-75	-75	-75	-75
Circ. Q (SUS)	-218	218	-218	218	-743	743	-725	725
Long. Pm (SUS)	288	288	288	288	288	288	288	288
Long. Pl (SUS)	-75	-75	-75	-75	-65	-65	-65	-65
Long. Q (SUS)	-736	736	-732	732	-220	220	-216	216
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	24	24	-24	-24	-24	-24	24	24
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	576.0	588.0	576.0	588.0	576.0	588.0	576.0	588.0
Pm+Pl (SUS)	512.9	524.8	512.9	524.8	503.1	515.0	503.1	515.0
Pm+Pl+Q (Total)	817	951	813	947	249	1256	235	1238

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	588.00	1407.13	Passed
Pm+Pl (SUS)	524.85	2110.70	Passed
Pm+Pl+Q (TOTAL)	1256.71	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	285	299	285	299	285	299	285	299
Circ. Pl (SUS)	-45	-45	-45	-45	-45	-45	-45	-45
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	142	142	142	142	142	142	142	142
Long. Pl (SUS)	-7	-7	-7	-7	-7	-7	-7	-7
Long. Q (SUS)	-285	285	-283	283	-287	287	-281	281
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	21	21	-21	-21	-21	-21	21	21
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	285.0	299.0	285.0	299.0	285.0	299.0	285.0	299.0
Pm+Pl (SUS)	244.0	257.6	244.0	257.6	244.0	257.6	244.0	257.6
Pm+Pl+Q (Total)	392.3	422.6	390.3	420.6	394.2	424.6	388.3	418.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	299.00	1203.20	Passed
Pm+Pl (SUS)	257.60	1804.80	Passed
Pm+Pl+Q (TOTAL)	424.59	3609.61	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: TS OUT Nozzle

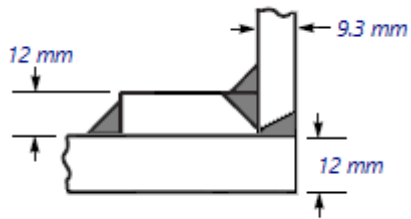
From: 50

Pressure for Reinforcement Calculations	P	9.232	kgf/cm ²
Temperature for Internal Pressure	Temp	125	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		854.35	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	7.1120	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	373.0500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS OUT Nozzle

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	10.020 in.
Actual Thickness Used in Calculation	0.365 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.23 \cdot 428.175 / (1407 \cdot 1.0 - 0.6 \cdot 9.23)$
 $= 2.8202 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.23 \cdot 130.43 / (1203 \cdot 1.0 - 0.6 \cdot 9.23)$
 $= 1.0054 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	521.7160 mm
Parallel to Vessel Wall, opening length	d	260.8580 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0625 mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

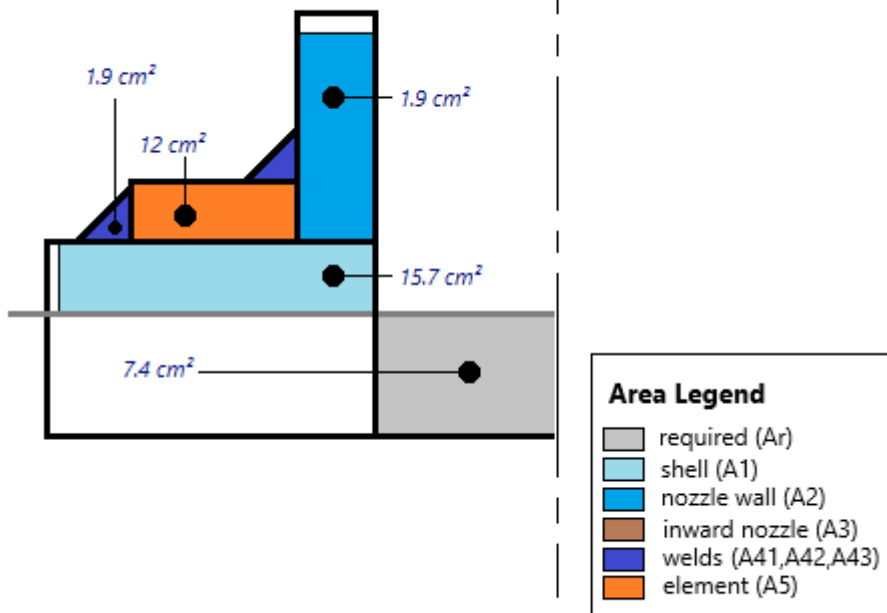


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7.357	NA	NA
Area in Shell	A1	15.664	NA	NA
Area in Nozzle Wall	A2	1.921	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	31.440	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (260.858 \cdot 2.8202 \cdot 1.0 + 2 \cdot 6.096 \cdot 2.8202 \cdot 1.0 \cdot (1 - 1.0))$
 $= 7.357 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 260.858(1.0 \cdot 8.825 - 1.0 \cdot 2.82) - 2 \cdot 6.096$
 $(1.0 \cdot 8.825 - 1.0 \cdot 2.8202) \cdot (1 - 1.0)$
 $= 15.664 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 22.06) \cdot (6.1 - 1.01) \cdot 0.8551$
 $= 1.921 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$



= 1.855 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (373.05 - 273.05) • 12.0 • 1.0
= 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 4.1804 mm
Wall Thickness per UG16(b), tr16b = 4.6750 mm
Wall Thickness, shell/head, internal pressure trb1 = 5.9952 mm
Wall Thickness tb1 = max(trb1, tr16b) = 5.9952 mm
Wall Thickness tb2 = max(trb2, tr16b) = 4.6750 mm
Wall Thickness per table UG-45 tb3 = 11.2776 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[11.278, max(5.9952, 4.675)]
= 5.9952 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(4.1804, 5.9952)
= 5.9952 mm

Available Nozzle Neck Thickness = 0.875 * 9.271 = 8.112 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	96.5,	Allowable	:	1203.2 kgf/cm ²	Passed
Expansion	:	0.0,	Allowable	:	2911.5 kgf/cm ²	Passed
Occasional	:	96.5,	Allowable	:	1600.3 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 8.112, tr = 1.005, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.204, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 8.112, tr = 1.005, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.204, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 8.112, tr = 1.005, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.204, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 2.82, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.32, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 8.112, tr = 1.005, c = 3.175 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.204, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 9.23/52.01 = 0.178

Weld Size Calculations, Description: TS OUT Nozzle

Intermediate Calc. for nozzle/shell Welds Tmin 6.0960 mm
 Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2672 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	4.4125 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = (Ar - A1) * Sv
 = (7.3567 - 15.664) * 1407
 = 0.00 kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²*fr2)*Sv
 = (1.9207 + 12.0 + 1.8551 - 0.0 * 0.86) * 1407
 = 22198.60 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 = (π/2) * Dlo * Wo * 0.49 * Snw
 = (3.1416/2.0) * 273.05 * 10.0 * 0.49 * 1203
 = 25287. kgf

Shear, Pad Element Weld [Spew]:
 = (π/2) * DP * WP * 0.49 * SEW
 = (3.1416/2.0) * 373.05 * 10.0 * 0.49 * 1407
 = 40403. kgf

Tension, Pad Groove Weld [Tpgw]:
 = (π/2) * Dlo * Wgpn * 0.74 * Seg
 = (3.1416/2) * 273.05 * 12.0 * 0.74 * 1407
 = 53593. kgf

Shear, Nozzle Groove Weld [Sngw]:
 = (π/2) * (Dlr+Wgnva) * (Wgnva-Can) * 0.60 * Sng
 = (3.14/2)*(260.858+7.112)*(7.112-3.175)*0.6*1407
 = 13991. kgf



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNGW) = (40403 + 13991) = 54395 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 54394 kgf, must exceed $W = 0 \text{ kgf}$ or $W1 = 22198 \text{ kgf}$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 16 kgf/cm^2

The Drop for this Nozzle is : 22.5252 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 284.5252 mm

Input Echo, WRC297 Item 1, Description: TS OUT Nozzle :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	850.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	254.508	mm
Nozzle Thickness	9.2710	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	373.050	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	9.23	kgf/cm ²
Radial Load	P	0.00	kgf
Circumferential Shear	Vc	0.00	kgf
Longitudinal Shear	Vl	0.00	kgf
Circumferential Moment	Mc	0.00	kgf-m.
Longitudinal Moment	Ml	0.00	kgf-m.
Torsional Moment	Mt	0.00	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(438.588(12.0 - 3.175))}$
 $= 478.355 \text{ mm}$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 877.175 mm
 Nozzle Outside Diameter (d) = 273.050 mm
 Vessel Thickness used (T) = 20.825 mm
 Nozzle Thickness used (t) = 6.096 mm
 T / t = 3.416
 d / t = 44.792

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.020$$

Nr/P = 0.048
 Mr/P = 0.059
 M0/P = 0.057
 N0/P = 0.201
 MrD/Mc = 0.132
 NrDL/Mc = 0.086
 M0d/Mc = 0.161
 N0DL/Mc = 0.296
 MrD/Ml = 0.078
 NrDL/Ml = 0.056
 M0D/Ml = 0.086
 N0DL/Ml = 0.408

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	185	194	185	194
Outplane Stress Summary	185	194	185	194

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	92	92	92	92
Inplane Stress Summary	92	92	92	92

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 185 194 185 194

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	92	92	92	92

Outplane Stress Summary 92 92 92 92

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	185	194	185	194

Inplane Stress Summary 185 194 185 194

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary 0 0 0 0

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress 185 194 185 194

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	231	242	231	242

Hoop Stress Summary 231 242 231 242

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	115	115	115	115

Axial Stress Summary	115	115	115	115
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	231	242	231	242

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	231	242	231	242

Hoop Stress Summary	231	242	231	242
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	115	115	115	115

Axial Stress Summary	115	115	115	115
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress 231 242 231 242

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(432.588(12.0 - 3.175))}$
 $= 476.946 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 865.175 mm
 Nozzle Outside Diameter (d) = 373.050 mm
 Vessel Thickness used (T) = 8.825 mm
 Nozzle Thickness used (t) = 56.096 mm
 $T / t = 0.157$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.269$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.048
 Mr/P = 0.078
 M0/P = 0.024
 N0/P = 0.043
 MrD/Mc = 0.251
 NrDL/Mc = 0.093
 M0d/Mc = 0.075
 N0DL/Mc = 0.073
 MrD/Ml = 0.062
 NrDL/Ml = 0.049
 M0D/Ml = 0.019
 N0DL/Ml = 0.044

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	443	452	443	452
Outplane Stress Summary	443	452	443	452

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	221	221	221	221
Inplane Stress Summary	221	221	221	221

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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longitudinal plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	0	0

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities			Top	Top	Bottom	Bottom
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			443	452	443	452

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses Normal to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Outplane Membrane (P)			0	0	0	0
Outplane Bending (P)			0	0	0	0
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			0	0	0	0
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			221	221	221	221
Outplane Stress Summary			221	221	221	221

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses parallel to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Inplane Membrane (P)			0	0	0	0
Inplane Bending (P)			0	0	0	0
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			0	0	0	0
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			443	452	443	452
Inplane Stress Summary			443	452	443	452

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to			Left	Left	Right	Right
circumferential plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			0	0	0	0

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			443	452	443	452

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int.	Stress Values at
---------------------	------------------

Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	185	194	185	194	185	194	185	194
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	92	92	92	92	92	92	92	92
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	185.0	194.0	185.0	194.0	185.0	194.0	185.0	194.0
Pm+Pl (SUS)	185.0	194.0	185.0	194.0	185.0	194.0	185.0	194.0
Pm+Pl+Q (Total)	185.0	194.0	185.0	194.0	185.0	194.0	185.0	194.0

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	194.00	1407.13	Passed
Pm+Pl (SUS)	194.00	2110.70	Passed
Pm+Pl+Q (TOTAL)	194.00	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	443	452	443	452	443	452	443	452
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	221	221	221	221	221	221	221	221
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	443.0	452.0	443.0	452.0	443.0	452.0	443.0	452.0
Pm+Pl (SUS)	443.0	452.0	443.0	452.0	443.0	452.0	443.0	452.0
Pm+Pl+Q (Total)	443.0	452.0	443.0	452.0	443.0	452.0	443.0	452.0

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	452.00	1407.13	Passed
Pm+Pl (SUS)	452.00	2110.70	Passed
Pm+Pl+Q (TOTAL)	452.00	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	231	242	231	242	231	242	231	242
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	115	115	115	115	115	115	115	115
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	231.0	242.0	231.0	242.0	231.0	242.0	231.0	242.0
Pm+Pl (SUS)	231.0	242.0	231.0	242.0	231.0	242.0	231.0	242.0
Pm+Pl+Q (Total)	231.0	242.0	231.0	242.0	231.0	242.0	231.0	242.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	242.00	1203.20	Passed
Pm+Pl (SUS)	242.00	1804.80	Passed
Pm+Pl+Q (TOTAL)	242.00	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description: Fixed TS

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	11.95 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	10.40 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00 kgf/cm ²
Shell Thickness	ts	12.0000 mm
Shell Internal Corrosion Allowance	cas	3.1750 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	850.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	125.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL1

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	9.23 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	4.59 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	12.0000 mm
Channel Internal Corrosion Allowance	cac	3.1750 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	850.000 mm
Channel Design Temperature	TEMPC	125.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	602
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	7309.00 mm
Straight Tube Length (bet. inner tubsht faces) L		7233.00 mm
Design Temperature of the Tubes		125.00 °C
Tube Material		SA-214
Tube Material UNS Number		K01807
Is this a Welded Tube		Yes
Tube Material Specification used		Wld. tube

Note:

Welded Tube Material Spec. already includes a Joint Eff. of 0.85. Since ASME checks Long. Tube stress, Jt. Eff. of long. seam is not applicable. Increasing the Tube allowables by dividing by 0.85.

Tube Allowable Stress at Temperature		801.45 kgf/cm ²
Tube Allowable Stress At Ambient		801.45 kgf/cm ²
Tube Allowable Stress at Temper.(for Pressure)		801.45 kgf/cm ²
Tube Allowable Stress at Temper. (for ASME)	St	942.89 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1641.67 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	23.8120 mm
Tube Layout Pattern		Square

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	419.210 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	38.0000 mm
Distance from Channel Side where Exp. Starts		0.0000 mm
Min. Distance from Shell Side where Exp. Stops		0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	3254.7 cm ²
Perimeter of Tube Layout	Cp	2309.1 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	125.00 °C
Tubesheet Material		SA-516 70

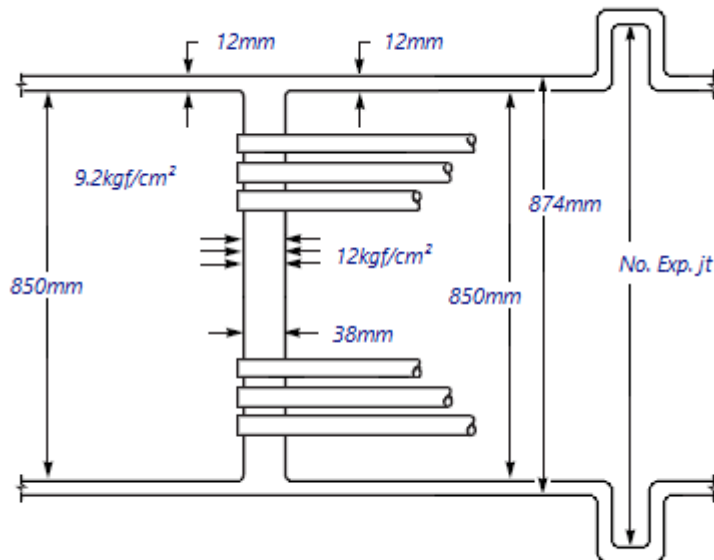
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K02700
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	38.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.1750 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.1750 mm
Tubesheet Outside Diameter	A	874.000 mm
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	727.500 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	62.76 °C
Mean Tube Metal Temp. along Tube length	Ttm	46.71 °C
Junction Stress Reduction option	Perform Plastic Calculation	
Tubesheet Integral with	Both	
Tubesheet Extended as Flange	No	
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No	

Tubesheet Integral w/Shell and Channel



Configuration a:

Messages regarding the Tubesheet & Exchanger Input Data:

*Warning - Length of Expanded portion of Tubes (38.000) is
 = Tubesheet Thickness (38.000 mm).*

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	62.8 °C	0.0000118609 /°C
Elastic Mod. at Design Temperature	125.0 °C	0.20048E+07 kgf/cm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	62.8 °C	0.20427E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	125.0 °C	0.20048E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	46.7 °C	0.0000117569 /°C
Elastic Mod. at Design Temperature	125.0 °C	0.20048E+07 kgf/cm²
Elastic Mod. at Metal Temp. along Len	46.7 °C	0.20521E+07 kgf/cm²
Elastic Mod. at Tubsht. Design Temp.	125.0 °C	0.20048E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	125.0 °C	0.20048E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²



Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (9.23 \cdot (19.05/2 - 0)) / (801.45 \cdot 1.0 + 0.4 \cdot 9.23)$
 $= 0.1092 + 0.0000 = 0.1092 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 125.00 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	7233.00	9.03	50.0000	0.0134948	977.27

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 153.1568 \text{ kgf/cm}^2$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3809	19.05	7233.00	50.01	50.0000	0.0004398	448.33

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 448.3348) / (3 \cdot 50.0125) = 11.9526 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3809 mm
Allowable Internal Pressure at Corroded thickness	194.80 kgf/cm ²
Required Internal Design Pressure	9.23 kgf/cm ²
Allowable External Pressure at Corroded thickness	153.16 kgf/cm ²
Required External Design Pressure	11.95 kgf/cm ²
Required Thickness due to Shell Side pressure	0.3809 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 38.0/38.0$
 $= 1.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$
 $= \max(19.05 - 2 \cdot 2.11 (2004811/2004811) \cdot (942/1407) \cdot (1.0); 19.05 - 2 \cdot 2.11)$
 $= 16.2223 \text{ mm}$

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 419.21 + 19.05$
 $= 857.470 \text{ mm}$

Determine the basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (23.812 - 19.05) / 23.812$
 $= 0.200$

Equivalent outer tube limit radius [ao]:

$= Do / 2$
 $= 857.4699/2$
 $= 428.7350 \text{ mm}$



Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 23.812 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 857.47 \cdot 23.812) / (\pi \cdot 857.47^2))}$$

$$= 23.8120 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (23.812 - 16.2223) / 23.812$$

$$= 0.3187$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 425.0 / 428.735$$

$$= 0.9913$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 425.0 / 428.735$$

$$= 0.9913$$

Compute parameter [xt]:

$$= 1 - Nt / (dt - 2 \cdot tt) / (2 \cdot a_o)^2$$

$$= 1 - 602 / ((19.05 - 2 \cdot 2.11) / (2 \cdot 428.735))^2$$

$$= 0.8199$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot a_o))^2$$

$$= 1 - 602 / (19.05 / (2 \cdot 428.735))^2$$

$$= 0.7029$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (D_s + ts) E_s / L$$

$$= \pi \cdot 12.0 (850.0 + 12.0) 2004811 / 7233.0$$

$$= 90072.6328 \text{ kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (D_t - tt) E_t / L$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 2004811 / 7233.0$$

$$= 311.2434 \text{ kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 90072.633 / (602 \cdot 311.243)$$

$$= 0.48072$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 90072.633 / 0)$$

$$= 1.0000 \text{ (= 1 if No Exp. Jt.)}$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((850.0 + 12.0) 12.0)$$

$$= 0.0179 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.018 \cdot 2004811 \cdot 12.0^3 / (6(1 - 0.3^2))$$

$$= 11340599.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



$$= 6(850.0 + 12.0) \cdot 11340599/38.0^3 \cdot (1 + 38.0 \cdot 0.018 + (38.0 \cdot 0.018)^2/2)$$

$$= 2041469.1250 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 850.0^2 / (4 \cdot 2004811 \cdot 12.0) \cdot (1 - 850.0 \cdot 0.3 / (2(850.0 + 12.0)))$$

$$= 0.63975\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / ((850.0 + 12.0)12.0))$$

$$= 0.0179 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.018 \cdot 2004811 \cdot 12.0^3 / (6(1 - 0.3^2))$$

$$= 11340599.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(850.0 + 12.0 / \cos(0)) \cdot 11340599/38.0^3 \cdot (1.0 + 38.0 \cdot 0.018 \cdot (1.0 + 1.0) / 2 + (38.0 \cdot 0.018)^2 / 2 \cdot 1.0)$$

$$= 2041469.1 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 850.0^2 / (4 \cdot 2004811 \cdot 12.0) \cdot (1 - 850.0 \cdot 0.3 / (2(850.0 + 12.0)))$$

$$= 0.63975\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 1.595834 ; \mu^* = 0.318735 ; E^* = 753607 \text{ kgf/cm}^2$$

$$E^*/E = 0.375899 ; \nu^* = 0.315982$$

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.316^2)602 \cdot 2004811 \cdot 2.11(19.05 - 2.11)428.735^2) / (753607 \cdot 7233.0 \cdot 38.0^3))^{1/4}$$

$$= 4.8920$$

Values from Table 4.18.3

$$Z_d = 0.013063 ; Z_v = 0.042708 ; Z_m = 0.300797$$

$$Z_a = 0.200760\text{E}+02 ; Z_w = 0.042708$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 874.0/857.4699$$

$$= 1.0193$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 753607 \cdot (2041469 + 2041469 + 2004811 \cdot \ln(1.02))$$

$$= 3.7407$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)^F$$

$$= (1 + 0.316)^{3.7407}$$

$$= 4.9226$$

Compute parameter [Q1]:



$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (0.9913 - 1 - 4.9226 \cdot 0.0427) / (1 + 4.9226 \cdot 0.3008)$$

$$= -0.088260$$

Compute Parameter [Qz1]:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.01306 + -0.08826 \cdot 0.04271) / 2 \cdot 4.89204^4$$

$$= 2.6613$$

Compute parameter [Qz2]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.04271 + -0.08826 \cdot 0.3008) / 2 \cdot 4.89204^4$$

$$= 4.6276$$

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.0427 + (0.9913 - 1) \cdot 0.3008) \cdot 4.89204^4 / (1 + 4.9226 \cdot 0.3008)$$

$$= 9.2553$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 0.9913 \cdot 11340599 \cdot 0.0179 \cdot 0.006397 (1 + 38.0 \cdot 0.0179)$$

$$= 2158.5220 \text{ mm}^2$$

Calculate parameter [w*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - w_s$$

$$= 428.735^2 (0.991^2 - 1) \cdot (0.991 - 1) / 4 - 2158.522$$

$$= -2151.5774 \text{ mm}^2$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C) -$$

$$\zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 0.991 \cdot 11340599 [0.006397 (1.0 + 0.018 \cdot 1.0 \cdot 38.0) -$$

$$0 (1.0/38.0 + 1.0 \cdot 0.018/2)]$$

$$= 2158.5220 \text{ mm}^2$$

Parameter [w*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - w_c$$

$$= 428.73495^2 [(0.99129^2 + 1) \cdot (0.99129 - 1) / 4 - (0.99129 - 1) / 2] - 2158.52197$$

$$= -2151.5774 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s -$$

$$[(p_s^2 - 1)/(J \cdot K_{st})] - [(1 - J)/(2J \cdot K_{st})][(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 11.952 \{ 0.703 + 2(1 - 0.703) \cdot 0.3 + [2/0.481 \cdot$$

$$(850.0/857.47)^2] \cdot 0.3 - [(0.991^2 - 1)/(1.0 \cdot 0.481)] -$$

$$[(1 - 1.0)/(2 \cdot 1.0 \cdot 0.481)] [(0^2 - (850.0)^2)/857.47^2] \}$$

$$= 25.6223 \text{ kgf/cm}^2$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.82 + 2(1 - 0.82) \cdot 0.3 + 1/(1.0 \cdot 0.481)] \cdot 9.232$$

$$= 27.7705 \text{ kgf/cm}^2$$



Pressure [Py]:

$$\begin{aligned} &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\ &= 602 \cdot 311.243 \cdot 0 / (3.142 \cdot 428.735^2) \\ &= 0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Pressure [Pw]:

$$\begin{aligned} &= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2) \\ &= -0 \cdot 9.255 \cdot 0 / (2 \cdot \pi \cdot 428.735^2) \\ &= -0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Calculate the pressure [Prim]:

$$\begin{aligned} &= -(U/ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) \\ &= -(9.255/428.735^2) \cdot (-2151.577 \cdot 11.952 - -2151.577 \cdot 9.232) \\ &= 0.2947 \text{ kgf/cm}^2 \end{aligned}$$

Calculate the pressure [Pw]:

$$\begin{aligned} &= (U/ao^2) \cdot (\omega \cdot s \cdot P^*s - \omega \cdot c \cdot P^*c) \\ &= (9.255/428.735^2) \cdot (2158.522 \cdot 0 - 2158.522 \cdot 0) \\ &= 0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot \\ &\quad (P^*s - P^*t + Py + Pw + Pw + Prim) \\ &= 0.1000E+01 \cdot 0.481 / (1 + 1.0 \cdot 0.481(2.661 + (0.991 - \\ &\quad 1)4.628)) \cdot (25.622 - 27.771 + 0 + 0 + 0 + 0.295) \\ &= -0.3943 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega \cdot s \cdot P^*s - \omega \cdot c \cdot P^*c)) + \\ &\quad W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\ &= [((-2151.577 \cdot 11.952 - -2151.577 \cdot 9.232) - \\ &\quad (2158.522 \cdot 0 - 2158.522 \cdot 0)) \cdot 0.01 + \\ &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.92264 \cdot 0.3008) \\ &= -23.5970 \text{ kgf} \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned} &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ &= -0.088 + 2 \cdot -23.597 / (-0.394 \cdot 428.735^2) \\ &= -0.02314 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.042185

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ &= (1.5 \cdot 0.0422 / 0.3187) \cdot (2 \cdot 428.735 / (38.0 - 0))^2 \cdot -0.39 \\ &= -39.8532 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 1407.13 = 2110.7 \text{ kgf/cm}^2$$

Note: Tubesheet Bending Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\ \text{Tubesheet Bending Stress} &= 524.6342 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\ &= (2 \cdot 0.2 \cdot 38.0 / 428.735) \cdot 1125.707 \\ &= 39.9065 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned} \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\ \text{Tubesheet Shear Stress} &= 32.7643 \text{ kgf/cm}^2 \end{aligned}$$



Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :
= max(HreqB; HreqS)
= max(3.81 ; 3.81)
= 3.8100 mm

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:
Ftmin = -0.5898, Ftmax = 3.4578

First Extreme Tube Axial Stress from among all the tubes [σ_1]:
= ((Ps • xs - Pt • xt) - Pe • Ftmin) / (Xt - Xs)
= ((11.95 • 0.7029 - 9.23 • 0.8199) - (-0.394) • -0.59) /
(0.8199 - 0.7029)
= 5.1175 kgf/cm²

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:
= ((Ps • xs - Pt • xt) - Pe • Ftmax) / (Xt - Xs)
= ((11.95 • 0.7029 - 9.23 • 0.8199) - (-0.394) • 3.458) /
(0.8199 - 0.7029)
= 18.7494 kgf/cm²

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
= max(| $\sigma_{t,1}$ | ; | $\sigma_{t,2}$ |) = 18.749 kgf/cm²

Allowable Tube Stress [σ_t ,A]:
= Sot = 942.8879 kgf/cm²

Largest tube-to-tubesheet Joint Load [Wt]:
= $\sigma_{t,max}$ • Tube Area
= 18.75 • 1.1229
= 21.05 kgf

Tube Metal Area [At]:
= $\pi (do - t) t$
= $\pi (19.05 - 2.11) 2.11$
= 1.123 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
= $\pi \cdot t (do - t) S$
= $\pi \cdot 2.11 (19.05 - 2.11) 942.888$
= 1058.779 kgf

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw$ (but not > Ft)
= $0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 942.89$
= 1058.7787 kgf

Weld Strength Factor [fw]:
= S / Sw where Sw = min(S and St)
= 942.888/942.888
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
= $0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 942.89$
= 703.4588 kgf

Minimum Length of Weld [ar]:
= $2 (\sqrt{ (0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd } - 0.75 \cdot do)$
= $2 (\sqrt{ (0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0 } - 0.75 \cdot 19.05)$
= 2.562 mm



Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
 Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1058.78 is >= tube strength 1058.78 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 1407.13 \cdot 1.0 = 1407.13 \text{ kgf/cm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 428.735^2 / ((850.0 + 12.0) 12.0) \cdot [-0.39 + (0.991^2 - 1)$
 $(11.95 - 9.23)] + 425.0^2 \cdot 9.23 / ((850.0 + 12.0) 12.0)$
 $= 153.3578 \text{ kgf/cm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 12.0 mm, the minimum Shell length adjacent to the tubesheet should be 181.791 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 428.735^2 / ((850.0 + 12.0) 12.0) [-0.39 + (0.991^2 - 1)$
 $(11.95 - 9.23)] + 425.0^2 \cdot 9.23 / ((850.0 + 12.0) 12.0)$
 $= 153.3578 \text{ kgf/cm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] +$
 $6(1 - \nu^2) / (E^*) (a_o/h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 11340599 / 12.0^2 \{ 0.018 [0.006397 \cdot 11.952 + 425.0^2$
 $\cdot 0 / (24057736)] +$
 $6(1 - 0.32^2) / (753607) (428.73 / 38.0)^3 (1 + 38.0 \cdot$
 $0.02 / 2) [-0.4 (0.043 + 0.301 \cdot -0.088) +$
 $2 / 428.73^2 \cdot 0.301 \cdot -23.597] \}$
 $= 553.9739 \text{ kgf/cm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$
 $:: |153.36| + |553.97| \leq 2110.7 \text{ kgf/cm}^2$
 $:: 707.33, \text{ must be } \leq 2110.7 \text{ kgf/cm}^2$

Note:
 For a given Channel thickness of 12.0 mm, the minimum Channel length adjacent to the tubesheet should be 181.791 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 425.0^2 \cdot 9.23 / ((850.0 + 12.0) 12.0 \cdot \cos(0))$
 $= 161.202 \text{ kgf/cm}^2$

Channel Bending Stress due to Joint Interaction [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t$
 $- 6(1 - \nu^2) / (E^*) \cdot (a_o/h)^3 \cdot (C_m \theta +$
 $h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 11340599 / 12.0^2 \cdot \{ (0.006397 \cdot 1.0 \cdot 0.018 - 0 \cdot 1.0) \cdot 9.232$
 $- 6(1 - 0.316^2) / 753607 \cdot$
 $(428.735 / 38.0)^3 \cdot (1.0 + 38.0 \cdot 0.018 / 2 \cdot 1.0) \cdot [-0.4 (0.04271 + 0.3008$
 $\cdot -0.088) + 2 / 428.735^2 \cdot 0.3008 \cdot -23.597] \}$
 $= 590.6269 \text{ kgf/cm}^2$

Channel Stress Summation vs. Allowable
 $:: |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$
 $:: |161.2| + |590.63| \leq 2110.7 \text{ kgf/cm}^2$
 $:: 751.83 \text{ must be } \leq 2110.7 \text{ kgf/cm}^2$



Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	39.9 <=	2110.7 kgf/cm ²	Pass
Tubesheet Shear Stress	7.3 <=	1125.7 kgf/cm ²	Pass
Maximum Tube Stress	18.7 <=	942.9 kgf/cm ²	Pass
Maximum Force on any one Tube	21.1 <=	1058.8 kgf	Pass
Axial Membrane Stress in Shell	153.4 <=	1407.1 kgf/cm ²	Pass
Shell Stress (jt. inter.)	707.3 <=	2110.7 kgf/cm ²	Pass
Channel Stress (jt. inter.)	751.8 <=	2110.7 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	3.810	38.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
			Bend	Allwd	Shear	Allwd		
D1uc	17.569	...	752	2111	114	1126	Ps+Pt-Th	D1 Ok
D2uc	16.881	...	741	2111	106	1126	Ps+Pt-Th	D2 Ok
D3uc	3.810	...	40	2111	7	1126	Ps+Pt-Th	D3 Ok
O1uc	15.550	...	1393	4792	235	1126	Ps+Pt+Th	O1 Ok
O2uc	5.124	...	374	4792	86	1126	Ps+Pt+Th	O2 Ok
O3uc	9.274	...	748	4792	143	1126	Ps+Pt+Th	O3 Ok
O4uc	11.840	...	1019	4792	179	1126	Ps+Pt+Th	O4 Ok
D1c	22.922	...	786	2111	132	1126	Ps+Pt-Th-c	D1 Ok
D2c	22.159	...	765	2111	120	1126	Ps+Pt-Th-c	D2 Ok
D3c	10.160	...	72	2111	13	1126	Ps+Pt-Th-c	D3 Ok
O1c	19.351	...	1210	4792	233	1126	Ps+Pt+Th-c	O1 Ok
O2c	10.160	...	330	4792	63	1126	Ps+Pt+Th-c	O2 Ok
O3c	13.766	...	548	4792	129	1126	Ps+Pt+Th-c	O3 Ok
O4c	16.049	...	819	4792	167	1126	Ps+Pt+Th-c	O4 Ok
Max:	22.9217	... mm		0.372		0.209	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures



Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	55	1407	...	...		...	...	...	...			Ok
D2uc	98	1407	...	...		...	...	...	...			Ok
D3uc	153	1407	...	...		...	...	...	...			Ok
O1uc	144	4792	-144	-1164		...	...	...	...			Ok
O2uc	86	4792	-86	-1164		...	...	...	...			Ok
O3uc	58	4792	-58	-1164		...	...	...	...			Ok
O4uc	171	4792	-171	-1164		...	...	...	...			Ok
D1c	79	1407	...	...		...	...	...	...			Ok
D2c	128	1407	...	...		...	...	...	...			Ok
D3c	208	1407	...	...		...	...	...	...			Ok
O1c	141	4792	-141	-1108		...	...	...	...			Ok
O2c	69	4792	-69	-1108		...	...	...	...			Ok
O3c	29	4792	-29	-1108		...	...	...	...			Ok
O4c	181	4792	-181	-1108		...	...	...	...			Ok
Max RATIO	0.148		0.163			...		...				

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Stress Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	86	942	-86	-502	96	1058	529	2110	1134	2110	Ok
D2uc	91	942	-47	-502	102	1058	1126	2110	382	2110	Ok
D3uc	19	942	...	...	21	1058	707	2110	752	2110	Ok
O1uc	252	1885	-79	-517	282	2117	1394	4792	1578	4792	Ok
O2uc	178	1885	...	...	200	2117	206	4792	682	4792	Ok
O3uc	211	1885	...	...	237	2117	414	4792	1246	4792	Ok
O4uc	219	1885	-36	-523	246	2117	1186	4792	1015	4792	Ok
D1c	88	942	-83	-576	99	1058	785	2110	1613	2110	Ok
D2c	88	942	-62	-561	99	1058	1577	2110	561	2110	Ok
D3c	26	942	...	...	29	1058	951	2110	1052	2110	Ok
O1c	251	1885	-65	-603	282	2117	1633	4792	1943	4792	Ok
O2c	153	1885	...	...	172	2117	188	4792	653	4792	Ok
O3c	197	1885	...	...	221	2117	261	4792	1455	4792	Ok
O4c	207	1885	-24	-614	232	2117	1322	4792	1141	4792	Ok
Max RATIO	0.133		0.171		0.133		0.747		0.764		

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	22.922	38.000	Ok
Tube thickness :	0.381	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Min Shell length of thk, (12.000) adj. to tubesheet: 181.791 mm

Min Channel length of thk, (12.000) adj. to tubesheet: 181.791 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.397 mm
 Due to Pressure Shell Compresses by : -0.051 mm
 Due to Pressure + Thermal Shell Compresses by : -1.442 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
--------------------------	---------------	-------------------------	----------------	------------------------



Tubesheet Bending Stress	25.927	0.996	34.428	1.000
Tubesheet Shear Stress	66.936	1.000	112.554	1.000
Tube Tensile Stress	98.634	1.000	123.861	1.000
Tube Compressive Stress	46.352	0.924	107.820	1.000
Tube-Tubesheet Joint load	98.634	1.000	123.861	1.000
Shell Stress (Axial, Junction)	25.927	0.996	34.236	0.994
Tubesheet-Channel Junction	20.902	1.000	34.236	0.994
Tube Pressure Stress	194.800	1.000	153.156	1.000
Minimum MAWP	20.902		34.236	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	25.927	0.998	34.236	1.000
Tubesheet Shear Stress	91.552	1.000	126.696	1.000
Tube Tensile Stress	101.368	1.000	123.861	1.000
Tube Compressive Stress	58.779	1.000	140.180	1.000
Tube-Tubesheet Joint load	101.368	1.000	123.861	1.000
Shell Stress (Axial, Junction)	25.927	0.998	34.236	1.000
Tubesheet-Channel Junction	25.927	0.998	34.236	1.000
Tube Pressure Stress	194.800	1.000	153.156	1.000
Minimum MAPnc	25.927		34.236	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed Delta Ms UnCorroded	Computed Delta Ms Corroded
1	0.0111	0.0111	...	...	...	...
2	-0.0511	-0.0511	...	...	...	...
3	-0.0399	-0.0400	...	...	...	...
4	...	...	...	...	...	...
5	0.0055	0.0055	-1.3962	-1.3974	...	...
6	-0.0444	-0.0445	-1.3962	-1.3974	...	...
7	-0.0389	-0.0389	-1.3962	-1.3974	...	...
8	...	...	-1.3962	-1.3974	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(38.0/4, \min(38.0, 12.0))$$

$$= 12.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(11.95/34.24, 1)$$

$$= 0.349$$

Thickness Ratio = 0.349, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(38.0/4, \min(38.0, 12.0))$$

$$= 12.000 \text{ mm}$$



MDMT reduction ratio per UCS 66 (b)(1)(b):
= min(pt/Tubeside MAPnc, 1)
= min(9.23/25.93, 1)
= 0.356

Thickness Ratio = 0.356, Temperature Reduction per Fig. UCS 66.1 = 73 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

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